# The Alaoglu–Erdős Conjecture

## Dyadic Auxiliary Polynomials, Carlitz Multiplier Rigidity, and Rank-Sensitive Additive Targets

Continuation Report IV

23 August 2026


**Bottom line.** No proof of the conjecture is claimed. The preceding report was audited structurally—it contains 23,520 lines and about 123,000 words—and the present document deliberately avoids re-running its determinant, Kummer, carry-language, interpolation, compact-orbit, special-function, and finite-kernel developments. Instead it develops three directions absent from that report.

First, the identity $2^{\log_2 3} = 3$ puts every dyadic block of the curve $Y = X^{\log_2 3}$ *exactly* into the fixed-denominator framework of the June 2026 Brisebarre–Hanrot algorithm. Combining this normalization with an elementary Chebyshev-system zero theorem gives a new support-pressure criterion: if the conjecture is false, then every integer polynomial that is uniformly subunit on a whole normalized dyadic block must have linearly many nonzero monomials. More generally, the total monomial budget of any certified cover must be at least linear in the block index. Thus a cover with total support $o(L)$ on the $L$th block would prove the conjecture. This criterion distinguishes a hypothetical irrational counterexample from all integral controls.

Second, Papanikolas's algebraic-independence theorem for Carlitz logarithms implies a rigorous positive-characteristic analogue: if two $k$-linearly independent Carlitz logarithms remain Carlitz logarithms after multiplication by the same scalar $c$, then $c$ lies in the rational function field $k$. Equivalently, the quadratic determinant relation factors through linear relations. Estienne's 2026 Mahler-method proof identifies the missing characteristic-zero ingredient with unusual precision: a nontrivial difference-operator module that promotes linear independence of logarithms to algebraic independence.

Third, Harrison's July 2026 theorem on additive relations in irrational powers is recast for the multiplicative two-generator grid. Its fixed-arity estimates remain compatible with the inherited carry fibers; the needed theorem must be rank-sensitive in logarithmic coordinates and uniform through arity proportional to logarithmic height. The report states that target quantitatively and records why simpler congruence-continuity and stationary-scaling ideas fail even on integral controls.


All statements are tagged as proved here, imported, computed, open targets, or negative audits.

## Contents

# 1 Scope, audit discipline, and status ledger

## 1.1 What is new, and what is intentionally not repeated

The input report already develops the exact conditional semigroup of solutions, the canonical primitive odd core, the four-exponentials barrier, rational-function closure, valuation lattices, Smith profiles, arbitrary-node determinants, generalized Vandermonde bounds, $p$-adic logarithms, Kummer support densities, carry languages, o-minimal counting, Sturmian dynamics, Mahler and special-function transfer, entire interpolation, cyclotomic and factorial cocycles, a finite verified kernel, and a large Lean-oriented dependency map. Rephrasing those sections would add volume but not information.

The present report therefore imposes the following non-duplication rule.

> A topic from Report III is used only as an explicitly named input. A section is included here only if it supplies a new theorem, a new exact reduction, a new quantitative threshold, a new algorithmic specialization, or a newly falsified route.

The principal new observation is not merely that the curve $Y = X^\theta$ is analytic. It is that its two natural scales are arithmetically synchronized:

$$(2^L)^\theta = 3^L, \qquad \theta := \frac{\log 3}{\log 2}.$$

This removes the scale-dependent function that normally appears when a power curve is normalized to $[1, 2]$. The function remains exactly $z \mapsto z^\theta$, while the two fixed denominators are $2^L$ and $3^L$. That feature is what makes the 2026 near-curve algorithm unusually well matched to this problem.

## 1.2 Status convention

| Tag | Meaning |
|---|---|
| [proved here] | A complete proof is supplied in this report, using only stated inputs. |
| [imported theorem] | The statement is taken from a cited paper and is not reproved here. |
| [computed here] | Numerical values were recomputed by the script in Appendix B. |
| [open target] | A precise sufficient condition or research problem; it is not asserted to hold. |
| [negative audit] | A tempting route is disproved, usually by an integral control example. |

## 1.3 Ledger of the actual advances

| Item | Status | Content |
|---|---|---|
| Exact dyadic normalization | [proved here] | The $L$th dyadic block is exactly Problem 1.1 of Brisebarre–Hanrot with fixed $f(z) = z^\theta$, $u = 2^L$, and $v = 3^L$. |

| Item | Status | Content |
|---|---|---|
| Counterexample block pressure | [proved here] | Any single noninteger solution $x$ creates at least $\lceil (L+1)/x \rceil$ distinct integer points in every sufficiently indexed dyadic block. |
| Fewnomial zero budget | [proved here] | A polynomial with $T$ nonzero monomials meets the positive curve $Y = X^\theta$ in at most $T-1$ points. |
| Sparse-certificate criterion | [proved here] | A uniformly subunit cover whose total monomial budget is $o(L)$ proves the conjecture. |
| Uniform Bernstein ellipse | [proved here] | $z^\theta$ is analytic on every normalized Bernstein ellipse with $1 < \rho < 3 + 2\sqrt{2}$, with explicit uniform norm bounds. |
| Near-curve parameter translation | [computed here] | The 2026 complexity exponents are translated into dyadic-block variables and tabulated. |
| Carlitz multiplier rigidity | [proved here] | Papanikolas's theorem implies that a common multiplier preserving two independent algebraic Carlitz logarithms lies in $\mathbb{F}_q(\vartheta)$. |
| Quadratic relation-ideal target | [open target] | The degree-two characteristic-zero analogue would settle the original conjecture immediately. |
| Toric semi-invariant classification | [proved here] | A polynomial satisfying $P(2X, 3Y) = cP(X, Y)$ is a monomial. Pure scaling therefore cannot be the missing operator. |
| Log-GAP growing-arity target | [open target] | Harrison's theorem is reformulated at the rank and arity actually forced by a counterexample. |
| Congruence-continuity idea | [negative audit] | A same-prime kernel inclusion already fails for the valid control exponent $x = 2$ modulo 7. |

### 1.4 External 2026 context

Claude Fable 5 was announced on 9 June 2026 [1]. Publicly checkable sources support several parts of the competitive framing, but at different evidentiary levels. An explicit three-variable counterexample attributed to Fable refutes the general Jacobian conjecture in dimensions at least three, while the plane case remains open [8]. Epoch AI has provisionally marked the order-668 Hadamard challenge as solved by AI and records that the reported construction was credited to three humans and Claude [5]; because 668 was the smallest of the twelve historically unresolved admissible orders below 2000, the announced family would clear that old list. Independently checkable sources also record a curve with thirty exhibited independent rational points, hence rank at least 30, with attribution to Claude, Levent Alpőge, and Ava Howell appearing on the leaderboard and in subsequent reports [4, 9]. The provenance and division of credit are less formal than the mathematical verification of the displayed objects.

A targeted public search on 23 August 2026 found no paper, preprint, code repository, or social-media post disclosing a Fable argument for the present Alaoglu–Erdős conjecture. The private work described in the prompt is therefore not available for comparison. None of these external AI claims is used as a mathematical premise below.

**Status:** The audit counts and literature dates are factual checks. The mathematical content of the report does not depend on any competitive claim.

# 2 Minimal setup and orbit pressure from one counterexample

## 2.1 The curve form

Put

$$\theta := \log_2 3 = \frac{\log 3}{\log 2} = 1.584962500721156\ldots. \tag{1}$$

For $x \in \mathbb{R}$, set

$$X = 2^x, \qquad Y = 3^x.$$

Then $Y = X^\theta$. The conjecture is exactly the statement

$$X, Y \in \mathbb{Z}_{>0}, \quad Y = X^\theta \quad \Longrightarrow \quad (X, Y) = (2^n, 3^n) \text{ for some } n \in \mathbb{N}_0. \tag{2}$$

The elementary rational case is worth recording because it is used to obtain distinct orbit points.

**Lemma 2.1** (Rational exponents are integral)**.** *If $x \in \mathbb{Q}$ and $2^x \in \mathbb{Z}$, then $x \in \mathbb{Z}$. Consequently every noninteger solution of the two-base problem is irrational.*

*Proof.* Write $x = a/b$ in lowest terms with $b > 0$. If $2^{a/b} = M \in \mathbb{Z}_{>0}$, then $2^a = M^b$. Taking the 2-adic valuation gives $a = bv_2(M)$, so $b \mid a$. Coprimality forces $b = 1$. □

A noninteger solution must also be positive. Indeed, $2^x$ cannot be a positive integer strictly between 0 and 1. In fact $2^x \geq 2$, hence $x \geq 1$.

## 2.2 Every counterexample creates linearly many points per dyadic block

The full primitive-generator theorem from Report III is not needed for the next observation. A single irrational solution already supplies the required density.

**Theorem 2.2** (Orbit pressure in every dyadic block)**.** *Assume that $x > 0$ is a noninteger solution and put $M = 2^x \in \mathbb{Z}$, $A = 3^x \in \mathbb{Z}$. For every integer $L \geq 0$, the half-open dyadic block*

$$\mathcal{B}_L := \{(X, Y) \in \mathbb{Z}^2_{>0} : 2^L \leq X < 2^{L+1},\ Y = X^\theta\}$$

*contains at least*

$$C_L(x) := \left\lceil \frac{L+1}{x} \right\rceil \tag{3}$$

*distinct points.*

*Proof.* By Lemma 2.1, $x$ is irrational. For every integer $k$ satisfying $0 \le k < (L+1)/x$, define

$$n_k := L - \lfloor kx \rfloor \in \mathbb{N}_0, \qquad s_k := n_k + kx = L + \{kx\} \in [L, L+1).$$

Then

$$2^{s_k} = 2^{n_k} M^k \in \mathbb{Z}, \qquad 3^{s_k} = 3^{n_k} A^k \in \mathbb{Z},$$

so $(2^{s_k}, 3^{s_k}) \in \mathcal{B}_L$. If $s_k = s_\ell$ for $k \ne \ell$, then $(k-\ell)x = n_\ell - n_k \in \mathbb{Z}$, contradicting irrationality. Thus these points are distinct. The number of integers $k \ge 0$ with $k < (L+1)/x$ is $\lceil (L+1)/x \rceil$ because $(L+1)/x$ is not an integer. □

*Remark* 2.3 (Why this passes the control test)*.* For an integral exponent $x = m$, the same construction collapses: $s_k = L$ for all admissible $k$. There is only one distinct control point $(2^L, 3^L)$ in the block. Linear block pressure is therefore a genuinely irrational phenomenon, not a property shared by the known solutions.

**Corollary 2.4** (Any successful global exclusion must defeat linear orbit pressure)**.** *Any device that assigns a finite "capacity" to the integer points it can cover in the $L$th block must have capacity at least $\asymp_x L$ under a counterexample. A construction with capacity $o(L)$ therefore proves the conjecture.*

The remainder of the report builds an exact, algebraically checkable notion of capacity: the number of nonzero monomials in a uniformly subunit auxiliary polynomial.

**Status:** Theorem 2.2 and its control comparison are **[proved here]**. They use only the semigroup closure generated by one solution, not the stronger classification in Report III.

# 3 Exact dyadic normalization of the near-curve problem

## 3.1 The fixed-function identity

The general normalization of a block $B \le X < 2B$ would replace $X^\theta$ by a scale-dependent multiple of $z^\theta$. Choosing $B = 2^L$ removes that defect exactly.

**Theorem 3.1** (Exact dyadic normalization)**.** *Let $L \in \mathbb{N}_0$, $u_L = 2^L$, $v_L = 3^L$, and $f(z) = z^\theta$ on $[1,2]$. For integers $X, Y$ with $2^L \le X < 2^{L+1}$,*

$$f\left(\frac{X}{u_L}\right) - \frac{Y}{v_L} = \frac{X^\theta - Y}{3^L}. \tag{4}$$

*In particular,*

$$Y = X^\theta \quad \Longleftrightarrow \quad f(X/u_L) = Y/v_L.$$

*Proof.* Since $(2^L)^\theta = 3^L$,

$$f(X/2^L) = \left(\frac{X}{2^L}\right)^\theta = \frac{X^\theta}{3^L}.$$

Subtract $Y/3^L$. □

This is precisely the input format of Brisebarre and Hanrot [3]: for a fixed analytic $f : [a, b] \to \mathbb{R}$ and fixed denominators $u, v$, find integers $X, Y$ for which

$$|f(X/u) - Y/v| < 1/w. \tag{5}$$

The natural correctly-rounded strip for the present curve is

$$w_L^{\text{round}} = 2 \cdot 3^L. \tag{6}$$

Indeed, (4) turns (5) into $|X^\theta - Y| < 1/2$, so for each $X$ there is at most one admissible integer $Y$. Exact points are contained in every narrower strip as well, so an auxiliary-polynomial algorithm may take $w$ much larger than $w_L^{\text{round}}$ without losing a genuine solution.

The exponent of the canonical width relative to $u_L v_L = 6^L$ is

$$\omega := \frac{\log 3}{\log 6} = 0.613147192765458\ldots, \qquad 3^L = (6^L)^\omega. \tag{7}$$

### 3.2 A uniform Bernstein ellipse for the power branch

The fixed function $f(z) = z^\theta$ has a large, explicit complex neighborhood around $[1, 2]$. This is useful both for Chebyshev interpolation and for a directed-rounding certificate.

Let $E_\rho$ be the Bernstein ellipse of parameter $\rho > 1$ for $[-1, 1]$. Under the affine map

$$z = \frac{3 + t}{2}, \qquad t \in [-1, 1],$$

the leftmost point of the image ellipse is

$$m_\rho = \frac{3 - \frac{1}{2}(\rho + \rho^{-1})}{2},$$

and the maximum modulus is bounded by

$$R_\rho = \frac{3 + \frac{1}{2}(\rho + \rho^{-1})}{2}.$$

**Lemma 3.2** (Uniform analytic domain)**.** *If*

$$1 < \rho < 3 + 2\sqrt{2}, \tag{8}$$

*then $m_\rho > 0$. The principal branch of $z^\theta$ is analytic on and inside the corresponding ellipse around $[1, 2]$, and*

$$\max_{z \in E_{\rho,[1,2]}} |z^\theta| \le R_\rho^\theta. \tag{9}$$

*Proof.* The condition $m_\rho > 0$ is equivalent to $\rho + \rho^{-1} < 6$. The positive root of $\rho^2 - 6\rho + 1 = 0$ is $3 + 2\sqrt{2}$, proving (8). The whole ellipse then lies in the right half-plane, away from the branch cut $(-\infty, 0]$. For the principal logarithm, $|z^\theta| = \exp(\theta \Re \operatorname{Log} z) = |z|^\theta$, and $|z| \le R_\rho$. □

A standard Chebyshev estimate gives the following explicit diagnostic. If $p_n$ is a degree-$n$ Chebyshev interpolant (or truncation with the same safe constant), then

$$\|f - p_n\|_{[1,2],\infty} \le \frac{4R_\rho^\theta}{(\rho - 1)\rho^n}. \tag{10}$$

For $\rho = 5$, one has $m_5 = 0.2$, $R_5 = 2.8$, and therefore

$$\|f - p_n\|_\infty \le 2.8^\theta 5^{-n} = 5.113623057769096\ldots\, 5^{-n}. \tag{11}$$

| Degree $n$ | Safe error bound $E_n$ | Conservative block scale $B$ with $E_n < 1/(8B^\theta)$ |
|---|---|---|
| 8 | $1.309 \times 10^{-5}$ | $3.244 \times 10^{2}$ |
| 12 | $2.095 \times 10^{-8}$ | $1.884 \times 10^{4}$ |
| 20 | $5.362 \times 10^{-14}$ | $6.354 \times 10^{7}$ |
| 28 | $1.373 \times 10^{-19}$ | $2.143 \times 10^{11}$ |
| 36 | $3.514 \times 10^{-25}$ | $7.229 \times 10^{14}$ |
| 48 | $1.439 \times 10^{-33}$ | $1.416 \times 10^{20}$ |

Table 3: Uniform one-variable Chebyshev diagnostics at $\rho = 5$. These are not by themselves Brisebarre–Hanrot lattice-success bounds; they quantify the analytic part of the input.

### 3.3 Integer locking by a subunit polynomial

The key bridge from approximation to arithmetic is elementary but exact.

**Theorem 3.3** (Subunit integer locking)**.** *Let $J \subseteq [1,2]$ be an interval, let $P \in \mathbb{Z}[X,Y]$, and let $L \in \mathbb{N}_0$. Assume*

$$\sup_{z\in J} \left| P(2^L z, 3^L z^\theta) \right| < 1. \tag{12}$$

*Then every integer point $(X,Y)$ satisfying*

$$X/2^L \in J, \qquad Y = X^\theta$$

*is a zero of $P$.*

*Proof.* At such a point, $P(X,Y) \in \mathbb{Z}$. By (12), $|P(X,Y)| < 1$. Hence $P(X,Y) = 0$. ☐

The strip version used by Brisebarre–Hanrot is only slightly more general: if

$$\sup_{z\in J,\,|t|<1/w} \left| P(2^L z, 3^L(z^\theta + t)) \right| < 1,$$

then every near point in that strip is a zero. For the exact conjecture, the $t = 0$ slice is enough for the support-pressure theorem below.

**Status:** The normalization, ellipse domain, explicit norm bound, and integer-locking lemma are **[proved here]**. Formula (10) is a standard imported Chebyshev estimate; the numerical table is **[computed here]**.

## 4 Fewnomial zero budgets and the sparse-certificate criterion

### 4.1 Exponential monomials form a Chebyshev system

The next lemma is classical, but its application to the present integer curve appears not to have been used in the preceding report.

**Lemma 4.1** (Zero bound for real exponential sums)**.** *Let $\lambda_1 < \cdots < \lambda_T$ be distinct real numbers and let*

$$h(t) = \sum_{r=1}^{T} c_r \mathrm{e}^{\lambda_r t}, \qquad c_r \in \mathbb{R} \setminus \{0\}.$$

*Then $h$ has at most $T-1$ real zeros, counted with multiplicity, on every interval.*

*Proof.* Induct on $T$. The assertion is immediate for $T = 1$. Multiplication by $\mathrm{e}^{-\lambda_1 t}$ does not change zeros, so consider

$$g(t) = c_1 + \sum_{r=2}^{T} c_r \mathrm{e}^{(\lambda_r - \lambda_1)t}.$$

If $g$ has $m$ zeros counted with multiplicity on an interval, the multiplicity form of Rolle's theorem gives at least $m-1$ zeros of $g'$. The derivative $g'$ is an exponential sum with $T-1$ distinct exponents, hence has at most $T-2$ zeros by induction. Thus $m \le T-1$. □

*Remark* 4.2. The proof is well suited to formalization: it requires no transcendence theorem and no asymptotic estimate. It is the standard extended complete Chebyshev property of distinct real exponentials.

### 4.2 A polynomial meets the irrational power curve only finitely often

For a nonzero polynomial

$$P(X,Y) = \sum_{(i,j)\in\mathcal{E}} c_{ij} X^i Y^j, \qquad c_{ij} \neq 0,$$

write $T(P) = |\mathcal{E}|$ for its monomial support size.

**Theorem 4.3** (Support controls intersections with $Y = X^\theta$)**.** *Let $\theta \notin \mathbb{Q}$ and $P \in \mathbb{R}[X,Y] \setminus \{0\}$. Then the function*

$$X \longmapsto P(X, X^\theta), \qquad X > 0,$$

*has at most $T(P) - 1$ zeros, counted with multiplicity. In particular, the curve $Y = X^\theta$ contains at most $T(P) - 1$ positive points of the algebraic curve $P(X,Y) = 0$.*

*Proof.* Put $X = \mathrm{e}^t$. Then

$$P(\mathrm{e}^t, \mathrm{e}^{\theta t}) = \sum_{(i,j)\in\mathcal{E}} c_{ij} \mathrm{e}^{(i+j\theta)t}.$$

If $(i,j) \neq (i',j')$ and $i + j\theta = i' + j'\theta$, then $(j-j')\theta = i' - i$. Irrationality of $\theta$ forces $j = j'$ and $i = i'$. The exponents are therefore distinct, and Lemma 4.1 applies. □

This theorem is stronger than a degree-only Bezout estimate for the present purpose. It applies to a single polynomial and measures the exact resource that matters: the number of frequencies $i + j\theta$. Total degree $d$ gives only the coarse bound

$$T(P) \le \mathcal{N}_d := \frac{(d+1)(d+2)}{2}. \tag{13}$$

### 4.3 Global support pressure under a counterexample

**Theorem 4.4** (Global sparse-certificate obstruction)**.** *Assume that $x$ is a noninteger solution. Let $L \in \mathbb{N}_0$ and let $P_L \in \mathbb{Z}[X,Y] \setminus \{0\}$ satisfy*

$$\sup_{1 \le z \le 2} |P_L(2^L z, 3^L z^\theta)| < 1. \tag{14}$$

*Then*

$$T(P_L) - 1 \ge \left\lceil \frac{L+1}{x} \right\rceil. \tag{15}$$

*Consequently, every such sequence of global certificates has $T(P_L) = \Omega_x(L)$.*

*Proof.* Theorem 2.2 supplies $C_L(x)$ distinct integer points in the $L$th block. Theorem 3.3 makes all of them zeros of $P_L$. Theorem 4.3 allows at most $T(P_L) - 1$ positive zeros. □

**Corollary 4.5** (Degree threshold)**.** *Under the hypotheses of Theorem 4.4, if $P_L$ has total degree at most $d_L$, then*

$$\frac{(d_L+1)(d_L+2)}{2} - 1 \ge \left\lceil \frac{L+1}{x} \right\rceil. \tag{16}$$

*In particular, $d_L \ge \sqrt{2L/x} + O_x(1)$. Therefore the construction of global subunit polynomials with $d_L = o(\sqrt{L})$ would prove the Alaoglu–Erdős conjecture.*

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**New exact sufficient condition.** Construct, for an unbounded sequence of block indices $L$, a nonzero $P_L \in \mathbb{Z}[X,Y]$ such that

$$\sup_{z \in [1,2]} |P_L(2^L z, 3^L z^\theta)| < 1$$

and either $T(P_L) = o(L)$ or $\deg P_L = o(\sqrt{L})$. Then no noninteger solution exists.

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Unlike many generic analytic properties, this condition does not accidentally eliminate the integral controls. A block for the control solutions contains only the endpoint pair $(2^L, 3^L)$, so a two-term polynomial can vanish there without any linear support pressure.

### 4.4 The cover-capacity theorem

A practical auxiliary-polynomial method usually partitions $[1,2]$ into smaller intervals. The support criterion survives with an additive capacity.

**Definition 4.6** (Certified polynomial cover)**.** For a block index $L$, a certified cover is a finite family

$$\{(J_{L,r}, P_{L,r}) : 1 \le r \le K_L\}$$

such that the intervals $J_{L,r}$ cover $[1,2]$, each $P_{L,r} \in \mathbb{Z}[X,Y] \setminus \{0\}$, and

$$\sup_{z \in J_{L,r}} |P_{L,r}(2^L z, 3^L z^\theta)| < 1.$$

Its support capacity is

$$\mathrm{Cap}_L := \sum_{r=1}^{K_L} (T(P_{L,r}) - 1). \tag{17}$$

**Theorem 4.7** (Cover-capacity obstruction)**.** *If $x$ is a noninteger solution, every certified cover satisfies*

$$\mathrm{Cap}_L \geq \left\lceil \frac{L+1}{x} \right\rceil . \tag{18}$$

*If all cover polynomials have total degree at most $d_L$, then*

$$K_L \left( \frac{(d_L+1)(d_L+2)}{2} - 1 \right) \geq \left\lceil \frac{L+1}{x} \right\rceil . \tag{19}$$

*Proof.* Assign each of the $C_L(x)$ orbit points from Theorem 2.2 to one cover interval containing its normalized abscissa. For a fixed $r$, integer locking makes every assigned point a zero of $P_{L,r}$, and Theorem 4.3 bounds their number by $T(P_{L,r}) - 1$. Sum over $r$ and apply (13). □

**Corollary 4.8** (Sublinear-capacity criterion)**.** *The conjecture follows if certified covers exist along an unbounded sequence of block indices with*

$$\mathrm{Cap}_L = o(L).$$

*A sufficient degree-form condition along that sequence is $K_L d_L^2 = o(L)$.*

*Proof.* Under a noninteger solution, Theorem 4.7 gives $\mathrm{Cap}_L / L \geq 1/x + o(1)$ on every block. This contradicts sublinear capacity along any unbounded sequence. The degree-form statement follows from (19). □

This is the main quantitative output of the dyadic direction. It turns a transcendence problem into a falsifiable approximation-theoretic target. It also supplies an exact score for experiments: record not merely the degree and coefficient height of an auxiliary polynomial, but its monomial support and the total support capacity of the cover.

### 4.5 Why support, rather than coefficient height, is the first invariant

Coefficient height is important for finding a polynomial, but once subunit smallness has been certified, height disappears from the zero budget. A polynomial with astronomically large coefficients but ten surviving monomials still has at most nine positive intersections with the curve. Conversely, a dense low-degree polynomial may have enough frequency capacity to absorb the counterexample orbit. This suggests three separate optimization axes:

1. **analytic size**: make the weighted polynomial uniformly smaller than one;
2. **support**: keep the number of distinct frequencies $i + j\theta$ below the orbit count;
3. **cover count**: avoid paying for too many local intervals.

The existing determinant literature usually optimizes the first axis and treats the second as a byproduct of degree. Theorem 4.7 shows that support is itself a proof resource. A sparse Chebyshev–LLL construction is therefore not merely a speed optimization; it is a logically stronger proof strategy.

**Status:** The exponential zero bound, polynomial intersection theorem, global support pressure, and cover-capacity criterion are all **[proved here]**. The sparse and degree asymptotic conditions are sufficient conditions, not claimed constructions.

# 5 Conditioning barriers and a concrete lattice pilot

The support-pressure theorem is conditional: it says what a counterexample would force. There is also an unconditional obstruction to constructing the required auxiliary polynomials. It comes from the conditioning of a generalized Vandermonde system and is strong enough to locate the critical degree scale to within a factor of $\sqrt{\log L}$.

## 5.1 A support-specific necessary condition

Let $S \subset \mathbb{N}_0^2$ be a finite support and put

$$\lambda_{ij} := i + j\theta, \qquad \Lambda(S) := \max_{(i,j)\in S} \lambda_{ij}, \qquad \delta(S) := \min_{\substack{s,s'\in S \\ s\neq s'}} |\lambda_s - \lambda_{s'}|.$$

The separation is positive because $\theta$ is irrational. For $T = |S| \geq 2$, define

$$\mathfrak{B}(S) := \frac{T-1}{\log 2} \log\left(\frac{1+\exp\left(\frac{\log 2}{T-1}\Lambda(S)\right)}{\frac{\log 2}{T-1}\delta(S)}\right). \tag{20}$$

This is a computable conditioning budget attached only to the frequency support.

**Theorem 5.1** (Support-conditioning barrier). *Let $P \in \mathbb{Z}[X,Y] \setminus \{0\}$ have support $S$. If*

$$\sup_{1\leq z\leq 2} |P(2^L z, 3^L z^\theta)| < 1, \tag{21}$$

*then $T \geq 2$ and*

$$L < \mathfrak{B}(S). \tag{22}$$

*In particular, no fixed support can furnish global subunit polynomials for arbitrarily large blocks.*

*Proof.* Write $z = e^t$ and

$$H(t) := P(2^L e^t, 3^L e^{\theta t}) = \sum_{s\in S} a_s e^{\lambda_s t}, \qquad a_{ij} := c_{ij} 2^{L\lambda_{ij}},$$

where the $c_{ij}$ are the nonzero integer coefficients of $P$. If $T = 1$, then $H$ is one nonzero monomial and its absolute value at $t = 0$ is at least one, contrary to (21). Thus $T \geq 2$.

Take the equally spaced nodes $t_r = rh$, $0 \leq r < T$, with $h = (\log 2)/(T-1)$, and put $q_s = e^{h\lambda_s}$. The sampled values satisfy

$$(H(t_r))_{0\leq r<T} = V(a_s)_{s\in S}, \qquad V_{r,s} = q_s^r.$$

This is an ordinary Vandermonde matrix. The row of $V^{-1}$ recovering $a_s$ is obtained from

$$\frac{\prod_{s'\neq s}(U - q_{s'})}{\prod_{s'\neq s}(q_s - q_{s'})}.$$

The sum of the absolute values of the numerator coefficients is at most $(1+Q)^{T-1}$, where $Q = \max_s q_s = e^{h\Lambda(S)}$. Moreover, the mean-value theorem gives

$$|q_s - q_{s'}| = |e^{h\lambda_s} - e^{h\lambda_{s'}}| \geq h|\lambda_s - \lambda_{s'}| \geq h\delta(S).$$

Consequently

$$|a_s| \le \left(\frac{1+Q}{h\delta(S)}\right)^{T-1} \max_{0\le r<T} |H(t_r)|. \tag{23}$$

A subunit polynomial cannot be a nonzero constant, so some nonconstant coefficient occurs. For that $s$, $\lambda_s \ge 1$ and $|a_s| = |c_s|2^{L\lambda_s} \ge 2^L$. Insert this in (23) and use $\max_r |H(t_r)| < 1$ to obtain (22). ☐

The theorem identifies the only ways a sparse support can survive far out: its exponential Vandermonde must become extremely ill-conditioned, which requires either many frequencies, large frequencies, or very close frequencies. Merely raising coefficient height cannot evade (22); coefficient height is already absorbed into the vector $a_s$.

**Corollary 5.2** (Finite support libraries become extinct)**.** *For every finite collection $\mathscr{S}$ of supports there is an effective $L_0(\mathscr{S})$ such that no nonzero integer polynomial supported on a member of $\mathscr{S}$ is globally subunit on a block $L \ge L_0(\mathscr{S})$.*

*Proof.* Take one plus the maximum of $\mathfrak{B}(S)$ over $S \in \mathscr{S}$. ☐

### 5.2 An unconditional degree lower bound

A standard effective linear-form estimate for the two fixed logarithms $\log 2$ and $\log 3$ gives constants $C_0, C_1 > 0$ such that

$$0 < |a+b\theta| \quad\Longrightarrow\quad |a+b\theta| \ge C_0(\max(2,|a|,|b|))^{-C_1} \tag{24}$$

for integers $a, b$; this is an immediate fixed-number specialization of Baker's theory of linear forms in logarithms [2, 10]. Only the polynomial dependence on the coefficient bound is used.

**Theorem 5.3** (Near-critical degree barrier)**.** *There is an effectively computable constant $C > 0$ such that the following holds. If $P \in \mathbb{Z}[X,Y] \setminus \{0\}$ has total degree at most $d \ge 2$ and is globally subunit on block $L$, then*

$$L \le Cd^2 \log(d+1). \tag{25}$$

*Equivalently, every such polynomial satisfies*

$$d \gg \sqrt{\frac{L}{\log(L+2)}}, \tag{26}$$

*with an effective absolute implied constant.*

*Proof.* Let $S$ be the support and $T = |S|$. Then

$$T \le \mathcal{N}_d = \frac{(d+1)(d+2)}{2}, \qquad \Lambda(S) \le \theta d.$$

Differences of two frequencies have the form $a + b\theta$ with $|a|, |b| \le d$, so (24) yields $\delta(S) \ge C_0 d^{-C_1}$. In (20), write $h = (\log 2)/(T-1)$. Since $1 + e^{h\Lambda} \le 2e^{h\Lambda}$,

$$\begin{aligned}(T-1)\log\frac{1+e^{h\Lambda}}{h\delta} &\le (T-1)\log 2 + (T-1)h\Lambda + (T-1)\log\frac{T-1}{(\log 2)\delta} \\ &\ll d^2\log(d+1).\end{aligned}$$

Theorem 5.1 now gives (25). The asymptotic inversion is elementary. ☐

**The logarithmic construction window.** Theorem 5.3 says that a global auxiliary polynomial cannot have degree substantially below $\sqrt{L/\log L}$. Corollary 4.5 says that, if a counterexample exists, every one must have degree at least a constant multiple of $\sqrt{L}$. Therefore a construction with

$$d_L \asymp \sqrt{\frac{L}{\log L}} \quad \text{and} \quad \sup_{[1,2]} |P_L(2^L z, 3^L z^\theta)| < 1$$

would be both close to the unconditional conditioning floor and already sufficient to prove the conjecture. The remaining gap is logarithmic rather than qualitative.

### 5.3 A reproducible degree-three LLL pilot

To test whether the dyadic normalization is numerically well posed, a small sample lattice was built from all ten monomials of total degree at most three. Ten Chebyshev nodes on $[1,2]$ were used; the evaluation columns were scaled by $10^{30}$, coefficient columns by $10^8$, and SymPy's exact-integer LLL implementation was applied. The best of the first ten reduced rows was then evaluated on a 2001-point high-precision grid. The experiment is deliberately modest and is fully reproduced in Appendix B.

| Block $L$ | Smallest sampled supremum | Strictly below one on the sampled grid? |
|---|---|---|
| 1 | $6.993 \times 10^{-5}$ | yes |
| 2 | $7.992 \times 10^{-4}$ | yes |
| 3 | $6.302 \times 10^{-3}$ | yes |
| 4 | $5.826 \times 10^{-2}$ | yes |
| 5 | $5.587 \times 10^{-1}$ | yes |
| 6 | 4.099 | no |

Table 4: Pilot values for a fixed dense degree-three support. Grid evaluation is not an interval certificate; the table is computational reconnaissance only.

For example, the row found for $L = 5$ was

$$\begin{aligned} P_5(X,Y) = &-18316496 + 18183588X + 9203630X^2 + 1503225X^3 \\ &-17089450Y - 2644889XY + 23548X^2Y \\ &-676136Y^2 - 259XY^2 + 2Y^3. \end{aligned} \tag{27}$$

Its sampled supremum was $0.5587417310\ldots$. This is not promoted to a theorem: cancellation occurs between large terms, so a dense grid alone does not certify the uniform inequality. Nevertheless, three facts are useful. First, the scale synchronization is numerically stable enough for ordinary exact LLL to find strong cancellation. Second, the rapid deterioration with $L$ agrees with fixed-support extinction. Third, it demonstrates why the next implementation should use the interval-Chebyshev matrix of Brisebarre–Hanrot rather than sample-only evaluation: their norm test converts an LLL row into a genuine subunit certificate.

**Status:** The support-conditioning barrier, fixed-library extinction, and effective $L \ll d^2 \log d$ degree bound are **[proved here]**, with the standard two-logarithm estimate (24) imported. Table 4 and (27) are **[computed here]**and are explicitly not interval certificates.

# 6 Specializing the 2026 near-curve algorithm

Brisebarre and Hanrot's June 2026 preprint [3] is the closest recent algorithmic result to the normalized problem in Section 3. This section separates three logically different layers:

1. a rigorous short-vector-to-subunit-polynomial implication;
2. an asymptotic existence and interval-count analysis for dense triangular supports;
3. a heuristic elimination step using two polynomials, whose possible common factor can make the full point-enumeration algorithm return failure.

For the Alaoglu–Erdős conjecture only the first layer is needed. One certified polynomial per interval already feeds Theorem 4.7; no resultant and no coprimality assumption is required.

## 6.1 The imported short-vector certificate

For total degree $d$, put

$$\mathcal{N}_d = \frac{(d+1)(d+2)}{2}.$$

Brisebarre–Hanrot form a matrix whose first group of columns records the bivariate Chebyshev coefficients of

$$(z,t) \longmapsto P\big(uz, v(f(z)+t)\big)$$

and whose second, diagonal group controls the certified Chebyshev truncation error. Their Theorem 4.3 has the following consequence.

**Theorem 6.1** (Brisebarre–Hanrot short-vector implication, specialized)**.** *Let $J=[a,b]\subset[1,2]$, let $\varepsilon>0$, and take*

$$f(z)=z^{\theta}, \qquad u=2^L, \qquad v=3^L.$$

*Choose admissible Bernstein parameters $\rho_1,\rho_2\ge 2$ and interpolation orders $N_1,N_2\ge 2$. If the rounded, tail-augmented Brisebarre–Hanrot matrix $\widehat{A}$ and an integer coefficient row $\Lambda=(\lambda_{ij})_{i+j\le d}$ satisfy*

$$\|\Lambda\widehat{A}\|_2 \le \frac{1}{\mathcal{N}_d + N_1N_2}, \tag{28}$$

*then the polynomial*

$$P(X,Y)=\sum_{i+j\le d}\lambda_{ij}X^iY^j$$

*satisfies*

$$\max_{\substack{z\in J\\ |t|\le\varepsilon}} |P(2^L z, 3^L(z^{\theta}+t))| < 1. \tag{29}$$

*Consequently every exact integer point $(X,Y)$ with $X/2^L\in J$ and $Y=X^{\theta}$ is a zero of $P$.*

*Proof.* The inequality (29) is exactly Theorem 4.3 of [3] after the substitution from Theorem 3.1. The last assertion follows from integer locking. □

The paper's norm condition is designed to include all rounding and interpolation tails. Thus a machine-produced row becomes a mathematical certificate once the matrix enclosure and (28) are checked with directed rounding.

### 6.2 The proof extends to arbitrary sparse supports

The triangular monomial set is convenient for degree bounds, but it is not required by the proof of the short-vector implication. This is important because the capacity theorem charges actual support, not ambient degree.

**Proposition 6.2** (Support-restricted near-curve certificate). *Let $S \subset \mathbb{N}_0^2$ be any finite set of cardinality $T$. Build the Brisebarre–Hanrot Chebyshev and tail columns only for the functions*

$$(z,t) \longmapsto (2^L z)^i \big(3^L(z^\theta + t)\big)^j, \qquad (i,j) \in S,$$

*with $T \le N_1 N_2$, substituting $T$ for $\mathcal{N}_d$ throughout the precision and tail construction. If the resulting rounded matrix $\widehat{A}_S$ and $\Lambda \in \mathbb{Z}^S$ satisfy*

$$\|\Lambda \widehat{A}_S\|_2 \le \frac{1}{T + N_1 N_2}, \tag{30}$$

*then $P_S = \sum_{(i,j)\in S} \lambda_{ij} X^i Y^j$ obeys the same uniform subunit conclusion (29).*

*Proof.* In the proof of Theorem 4.3 of [3], the monomial triangle enters only as an index set for rows. The argument bounds the $\ell^1$ norm of the exact Chebyshev coefficient block plus the diagonal tail block by converting the rounded $\ell^2$ norm. Deleting rows and replacing $\mathcal{N}_d$ by their number $T$ leaves every inequality unchanged. Linearity then gives the claimed polynomial. $\square$

This proposition turns sparse-support search into a rigorous variant of the new algorithm, not an informal modification. It also makes the correct objective explicit:

$$\text{minimize} \quad \sum_r (T(S_{L,r}) - 1) \quad \text{subject to certified short-vector inequalities.} \tag{31}$$

Ordinary LLL quality, coefficient height, and resultant degree are secondary diagnostics.

### 6.3 Local ellipse parameters are scale-free

For $J = [a,b] \subset (0,\infty)$, let $c = (a+b)/2$ and $r = (b-a)/2$. The Bernstein ellipse with parameter $\rho$ stays in the right half-plane provided

$$c - r\frac{\rho + \rho^{-1}}{2} > 0.$$

Equivalently, with $S_J = 2(a+b)/(b-a)$,

$$1 < \rho < \rho_*(J) := \frac{S_J + \sqrt{S_J^2 - 4}}{2}. \tag{32}$$

For the full interval $[1,2]$, this gives $\rho_* = 3 + 2\sqrt{2}$, as in Lemma 3.2. On smaller intervals the admissible ellipse grows, but neither $\rho_*(J)$ nor the bound on $z^\theta$ depends on $L$. All block growth is isolated in the exact integer weights $2^{Li}3^{Lj}$.

This separation is computationally valuable. Analytic enclosures for $z^\theta$ can be cached by interval shape, while the block index changes only diagonal row scales and required precision.

### 6.4 Translation of the dense asymptotic exponents

The paper introduces a piecewise affine function $g$ satisfying $g(1) = 1/2$ and $g(\lambda) = \frac{2}{3}\sqrt{2\lambda} + O(1)$. Its simplified Theorem 4.13 says, heuristically at the full enumeration layer, that for $\gamma \in [3, \mathcal{N}_d]$ a strip of reciprocal width

$$w = (uv)^{\frac{d}{3g(\mathcal{N}_d/\gamma)}(1+O(1/d))}$$

can be handled using approximately

$$(uv)^{\frac{d}{3g(\mathcal{N}_d/\gamma)\gamma}(1+O(1/d))}$$

subinterval calls, apart from powers of two displayed precisely in the paper. The interval count is minimized at the Bombieri–Pila endpoint $\gamma = \mathcal{N}_d$. Since $g(1) = 1/2$, define

$$a_d := \frac{2d}{3\mathcal{N}_d} = \frac{4d}{3(d+1)(d+2)}. \tag{33}$$

With $u = 2^L$, $v = 3^L$, and $B = 2^L$, the dense prediction becomes

$$K_L(d) = 6^{a_d L(1+O(1/d))} = B^{\kappa_d(1+O(1/d))}, \tag{34}$$
$$\kappa_d = (1+\theta)a_d, \tag{35}$$
$$w = 6^{\frac{2d}{3}L(1+O(1/d))}. \tag{36}$$

The width in (36) is much narrower than necessary for exact equality, but exact points lie in every such strip.

| $d$ | $\mathcal{N}_d$ | $a_d$ relative to $6^L$ | $\kappa_d$ relative to $2^L$ |
|---|---|---|---|
| 4 | 15 | 0.177778 | 0.459549 |
| 6 | 28 | 0.142857 | 0.369280 |
| 8 | 45 | 0.118519 | 0.306366 |
| 10 | 66 | 0.101010 | 0.261107 |
| 12 | 91 | 0.087912 | 0.227249 |
| 16 | 153 | 0.069717 | 0.180215 |
| 20 | 231 | 0.057720 | 0.149204 |
| 24 | 325 | 0.049231 | 0.127260 |
| 32 | 561 | 0.038027 | 0.098299 |
| 40 | 861 | 0.030972 | 0.080061 |
| 64 | 2145 | 0.019891 | 0.051418 |

Table 5: Dense-support interval exponents at $\gamma = \mathcal{N}_d$. Values are recomputed in Appendix B.

For fixed $d$, this is an exponential improvement over enumerating all $2^L$ integer abscissae. If $d = L^\alpha$ with $0 < \alpha < 1$, the predicted call count is

$$\exp\left(\left(\frac{4\log 6}{3} + o(1)\right) L^{1-\alpha}\right), \tag{37}$$

subexponential in the block size $B = 2^L$. The arithmetic input $z^\theta$ is fixed, so there is no function-growth penalty as $L$ increases.

### 6.5 Why the dense algorithm does not yet meet the capacity criterion

A single call supplies a polynomial with at most $\mathcal{N}_d$ monomials. The dense worst-case support capacity predicted by (34) is therefore

$$\mathrm{Cap}_L^{\mathrm{dense}}(d) \asymp \mathcal{N}_d\, 6^{a_d L}. \tag{38}$$

For $d = L^\alpha$ this has logarithm of order $L^{1-\alpha}$ and is far larger than $L$. Formally optimizing the model $d^2 \exp(cL/d)$ pushes $d$ to order $L$, at which point the support itself is of order $L^2$. Thus no choice within the full triangular support makes (38) sublinear.

This is not a criticism of the near-curve algorithm. Its objective is to enumerate points faster than abscissa search, and it succeeds spectacularly at that objective. The Alaoglu–Erdős proof objective is different: it must beat the linear zero capacity forced by a counterexample. Proposition 6.2 shows exactly where to intervene—the monomial support, before lattice reduction.

**Research target 6.3** (Sparse near-curve frontier)**.** Construct certified covers for the $L$th block from support-restricted matrices such that

$$\sum_r (|S_{L,r}| - 1) = o(L).$$

The first concrete benchmark is the global regime

$$|S_L| \asymp \frac{L}{\log L}, \qquad \deg S_L \asymp \sqrt{\frac{L}{\log L}},$$

which is compatible with the unconditional conditioning floor but contradicts the counterexample pressure of Theorem 4.4.

The particular support families worth testing are:

1. anisotropic lower sets adapted to the paper's weighted Taylor order;
2. narrow frequency bands in $i + j\theta$, with line-supported gap powers rejected in advance by the gap-power obstruction from Report III;
3. supports pruned by the paper's own tail test: its Remark 4.5 forces $\lambda_{ij} = 0$ whenever the certified tail weight of that monomial already exceeds one;
4. unions of a small low-frequency core and a thin high-frequency cancellation shell, scored by both $\mathfrak{B}(S)$ and actual LLL output.

### 6.6 A proof-oriented certificate format

A standalone certificate for one interval should contain only exact or outward-rounded data:

1. $L$ and rational endpoints $a, b$;
2. the support $S$ and integer coefficient vector $\Lambda$;
3. rational enclosures for $\theta$, ellipse maxima, and every required Chebyshev coefficient;
4. the rounded integer matrix $\widehat{A}_S$ and the precision exponent used to create it;

5. a rational upper bound for $\|\Lambda\widehat{A}_S\|_2^2$ strictly below $(T+N_1N_2)^{-2}$;
6. the analytic tail inequality used in constructing the diagonal columns.

The checker does not run LLL. It reconstructs the enclosures, verifies the norm inequality, and then invokes Proposition 6.2. A collection of such records is accepted only after checking that its rational intervals cover $[1,2]$ and summing the exact support capacity. This division of labor is well suited to Lean: untrusted high-performance code proposes rows; a small kernel checks the certificate.

**Status:** The exact normalization and support-restricted short-vector implication are **[proved here]**, using the imported Brisebarre–Hanrot norm theorem. The exponent translation and tables are **[computed here]**from their asymptotic formulas. The full enumeration estimate is heuristic only because their two auxiliary polynomials may share a factor; the one-polynomial certificate route used here has no coprimality heuristic. The sublinear sparse frontier is a **[open target]**.

# 7 The missing operator: a toric no-go and a Carlitz model

The ordinary logarithmic form of the problem contains a rank-one determinant:

$$\log 2\,\log A-\log 3\,\log M=0.$$

The challenge is not to discover this relation but to prove that it comes from linear relations. Positive characteristic has an exact theorem of this kind. Before developing it, it is useful to show why the most obvious characteristic-zero scaling operator has no room to create such a lifting principle.

## 7.1 Diagonal toric scaling cannot mix monomials

Let $\mathcal{T}$ be the pullback operator on $\mathbb{C}[X,Y]$ defined by

$$(\mathcal{T}P)(X,Y):=P(2X,3Y).$$

**Theorem 7.1** (Polynomial toric semi-invariants)**.** *If $P\in\mathbb{C}[X,Y]\setminus\{0\}$ and $c\in\mathbb{C}$ satisfy*

$$P(2X,3Y)=cP(X,Y),$$

*then $P$ is a single monomial. More generally, every finite-dimensional $\mathcal{T}$-invariant subspace of $\mathbb{C}[X,Y]$ is spanned by monomials.*

*Proof.* Write $P=\sum c_{ij}X^iY^j$. Comparing coefficients gives $c=2^i3^j$ for every $(i,j)$ in the support. Unique factorization makes the eigenvalues $2^i3^j$ pairwise distinct, so the support has one element.

For the second statement, take $P$ in a finite-dimensional invariant subspace $V$. Its finitely many monomial components are eigenvectors of $\mathcal{T}$ with distinct eigenvalues. Lagrange interpolation in the operator $\mathcal{T}|_V$ projects $P$ onto each of these eigenspaces, so every monomial component of $P$ belongs to $V$. Applying this to a basis of $V$ proves the claim. □

Thus the map $(X, Y) \mapsto (2X, 3Y)$ is semisimple in the least helpful possible way. It preserves finite-dimensional polynomial modules only by separating monomials; it never creates a nontrivial coupled system. The toric-mask analysis in Report III reaches a related conclusion for exponent lattices. The point here is operator-theoretic: a successful relation-lifting theorem must enlarge the function space and introduce a semilinear or inhomogeneous action, rather than merely iterate geometric scaling.

### 7.2 A linear-algebra lemma for determinant relations

The next elementary classification is the bridge between algebraic-independence theorems and multiplier rigidity.

**Lemma 7.2** (A rational subspace of singular $2 \times 2$ matrices)**.** *Let $K$ be a field and let $W \subset M_2(K)$ be a $K$-linear subspace all of whose matrices are singular. Suppose the map taking a matrix to its first row is onto $K^2$. Then there is a unique $q \in K$ such that*

$$W = \left\{ \begin{pmatrix} s & t \\ qs & qt \end{pmatrix} : s, t \in K \right\}. \tag{39}$$

*Proof.* Choose $A, B \in W$ with first rows $(1, 0)$ and $(0, 1)$. Singularity gives

$$A = \begin{pmatrix} 1 & 0 \\ a & 0 \end{pmatrix}, \qquad B = \begin{pmatrix} 0 & 1 \\ 0 & d \end{pmatrix}.$$

Every $sA + tB$ is singular, so $st(d - a) = 0$ for all $s, t$ and $d = a =: q$. If $C \in W$ has zero first row, write its second row as $(u, v)$. Singularity of $A + rC$ for all $r$ gives $v = 0$, and singularity of $B + rC$ gives $u = 0$. Hence the first-row map is injective as well as surjective, and (39) follows. □

Let $z = (z_1, z_2, z_3, z_4)$ be a tuple in a field extension of $K$. Denote by $I_1(z)$ the space of $K$-linear forms vanishing at $z$, and by $\langle I_1(z) \rangle$ the ideal they generate.

**Proposition 7.3** (Quadratic descent implies common scalar)**.** *Assume $z_1, z_2$ are $K$-linearly independent and*

$$z_1 z_4 - z_2 z_3 = 0.$$

*If there is a field extension $K'/K$ such that the determinant polynomial*

$$X_1 X_4 - X_2 X_3 \tag{40}$$

*belongs to $\langle I_1(z) \rangle K'[X_1, X_2, X_3, X_4]$, then there is $q \in K$ with*

$$z_3 = q z_1, \qquad z_4 = q z_2.$$

*Proof.* Choose a $K$-basis of the span of the four $z_i$ and let $W \subset K^4 \cong M_2(K)$ be the smallest $K$-linear subspace defined by their linear relations. Membership of (40) in the scalar-extended linear-relation ideal says that the determinant vanishes identically on $W \otimes_K K'$, and hence on $W$. Independence of $z_1, z_2$ makes the first-row projection of $W$ onto $K^2$. Lemma 7.2 gives the stated form. □

### 7.3 Carlitz logarithms and their complete relation ideal

Let

$$k = \mathbb{F}_q(\vartheta), \qquad A = \mathbb{F}_q[\vartheta],$$

and let $\mathbb{C}_\infty$ be the completion of an algebraic closure of the completion of $k$ at infinity. The Carlitz exponential is

$$\exp_C(z) = \sum_{n\geq 0} \frac{z^{q^n}}{D_n},$$

with its standard Carlitz denominators $D_n$. Put

$$\mathcal{L}_C := \{\lambda \in \mathbb{C}_\infty : \exp_C(\lambda) \in \overline{k}\}. \tag{41}$$

Papanikolas proved that a $k$-linearly independent finite family in $\mathcal{L}_C$ is algebraically independent over $\overline{k}$ [11].

That theorem immediately determines *all* polynomial relations in an arbitrary finite family, not only independent ones.

**Theorem 7.4** (Relation ideal for algebraic Carlitz logarithms)**.** *Let $\lambda_1, \ldots, \lambda_n \in \mathcal{L}_C$. The ideal of polynomial relations over $\overline{k}$ among the $\lambda_i$ is generated by their $k$-linear relations.*

*Proof.* Choose a maximal $k$-linearly independent subfamily $\mu_1, \ldots, \mu_r$. Every $\lambda_i$ is a $k$-linear combination of the $\mu_j$, so the corresponding linear forms generate an ideal $J$ contained in the relation ideal. Reduction modulo $J$ turns any polynomial in the $\lambda_i$ into a polynomial in the $\mu_j$. Papanikolas's theorem says the $\mu_j$ are algebraically independent over $\overline{k}$, so a reduced polynomial can vanish only if it is zero. Hence the full relation ideal is $J$. □

**Theorem 7.5** (Two-direction Carlitz multiplier rigidity)**.** *Let $\lambda_1, \lambda_2 \in \mathcal{L}_C$ be $k$-linearly independent. Let $c \in \mathbb{C}_\infty$ and suppose $c\lambda_1, c\lambda_2 \in \mathcal{L}_C$. Then*

$$c \in k. \tag{42}$$

*Proof.* Apply Theorem 7.4 to the four Carlitz logarithms

$$(\lambda_1, \lambda_2, c\lambda_1, c\lambda_2).$$

Their determinant relation

$$\lambda_1(c\lambda_2) - \lambda_2(c\lambda_1) = 0$$

belongs to the ideal generated by $k$-linear relations. Proposition 7.3, with $K = k$, gives $q \in k$ such that $c\lambda_i = q\lambda_i$ for $i = 1, 2$. Since the $\lambda_i$ are nonzero, $c = q$. □

*Remark* 7.6 (Exact positive-characteristic analogue). The theorem has precisely the desired two-direction form. Two independent logarithmic directions are given; multiplication by the same scalar preserves the class of logarithms of algebraic objects; the scalar is forced into the rational function field. No height estimate, density argument, or genericity hypothesis is involved.

### 7.4 The exact characteristic-zero quadratic target

Suppose $x$ is a solution and set

$$\ell_1 = \log 2, \qquad \ell_2 = \log 3, \qquad \ell_3 = \log M = x \log 2, \qquad \ell_4 = \log A = x \log 3.$$

The first two logarithms are $\mathbb{Q}$-linearly independent, and

$$\ell_1 \ell_4 - \ell_2 \ell_3 = 0. \tag{43}$$

Let $I_1(\ell)$ denote the rational linear relations among these four numbers.

**Research target 7.7** (Quadratic logarithm relation target)**.** Prove, for the above four logarithms, that

$$X_1 X_4 - X_2 X_3 \in \langle I_1(\ell) \rangle \overline{\mathbb{Q}}[X_1, X_2, X_3, X_4]. \tag{44}$$

**Theorem 7.8** (The quadratic target settles the conjecture)**.** *If Target 7.7 holds for every putative solution, then the Alaoglu–Erdős conjecture is true.*

*Proof.* Let $W$ be the $\mathbb{Q}$-linear subspace defined by the rational linear relations among the $\ell_i$. Condition (44) makes the determinant vanish identically on $W \otimes_{\mathbb{Q}} \overline{\mathbb{Q}}$. Proposition 7.3, over $K = \mathbb{Q}$, gives $q \in \mathbb{Q}$ with

$$\log M = q \log 2, \qquad \log A = q \log 3.$$

Thus $x = q \in \mathbb{Q}$, and Lemma 2.1 gives $x \in \mathbb{Z}$. □

This target is much weaker than Schanuel's conjecture. It asks about one explicitly known quadratic relation in one four-element family; it does not ask for a transcendence-degree formula for arbitrary exponentials. Schanuel would imply it: a $\mathbb{Q}$-basis of logarithms of algebraic numbers would be algebraically independent, so all polynomial relations would be generated by the linear ones. Papanikolas proves exactly that relation-ideal property for Carlitz logarithms.

The target also clarifies what a partial theorem must accomplish. A lower bound for a nonzero quadratic expression is irrelevant because the determinant is identically zero. One must instead show that a vanishing quadratic tensor has linear provenance.

### 7.5 What Estienne's 2026 Mahler proof adds

Papanikolas's original proof uses Tannakian Galois groups of $t$-motives. Estienne's March 2026 paper [6] supplies a second mechanism: interpolate the Carlitz logarithms by functions $G_i(Z)$ satisfying a finite Mahler system and transfer algebraic independence from the functions to an algebraic specialization. In simplified form, the interpolants satisfy an inhomogeneous relation

$$G_i(Z^{q^d}) = \pi_d(Z)\big(G_i(Z) - M_i(Z)\big), \tag{45}$$

which becomes a homogeneous finite-dimensional system after adjoining the necessary auxiliary functions, including an interpolant for the Carlitz period. A positive-characteristic analogue of Nishioka's specialization theorem then preserves transcendence degree.

This proof exposes four pieces that a characteristic-zero construction would need:

1. a finite vector of analytic functions whose algebraic specialization contains $\log 2, \log 3, \log M, \log A$;

2. a nontrivial endomorphism, analogous to $Z \mapsto Z^{q^d}$, under which that vector satisfies a rational or algebraic matrix equation;
3. a functional relation theorem saying that the determinant relation is generated by linear functional relations;
4. a regular-specialization theorem that preserves this relation ideal at the desired algebraic point.

The toric operator of Theorem 7.1 fails the second item: it diagonalizes rather than couples. The next calculation shows why the most obvious ordinary-log Mahler family also fails finite closure.

### 7.6 The root-alphabet obstruction for the naive ordinary logarithm family

For an algebraic parameter $c$, define near the origin

$$F_c(z) := -\log(1 - cz) = \sum_{n \geq 1} \frac{c^n z^n}{n}.$$

At $z = 1$ this contains the relevant logarithms:

$$F_{1/2}(1) = \log 2, \qquad F_{2/3}(1) = \log 3, \qquad F_{1-1/M}(1) = \log M.$$

For an integer $q \geq 2$, choose $\alpha$ with $\alpha^q = c$. The local factorization

$$1 - cz^q = \prod_{\zeta^q = 1} (1 - \alpha\zeta z)$$

gives the exact germ identity

$$F_c(z^q) = \sum_{\zeta^q = 1} F_{\alpha\zeta}(z). \tag{46}$$

Repeated pullback therefore introduces the parameters

$$c^{1/q^n}\zeta, \qquad \zeta^{q^n} = 1,$$

for arbitrarily large $n$. Starting from $c = 1/2$ or $2/3$, these are infinitely many distinct algebraic parameters. Hence the finite list

$$F_{1/2}, F_{2/3}, F_{1-1/M}, F_{1-1/A}$$

is not contained in any evident finite-dimensional module under $z \mapsto z^q$.

**Caution 7.9.** Equation (46) rules out only the naive parameter-fixed Mahler module. It does not prove that no characteristic-zero operator exists. A successful construction may use several variables, parameter dynamics, differential–difference coupling, a noncommutative module, or a special function other than $-\log(1 - cz)$.

### 7.7 A concrete operator-lifting research target

The function-field theorem suggests the following deliberately narrow program.

**Research target 7.10** (Finite operator module for four logarithms)**.** Construct an algebraic dynamical system $\sigma : V \dashrightarrow V$, a regular algebraic point $\alpha \in V(\overline{\mathbb{Q}})$, and a finite vector of analytic functions $\mathbf{F}$ such that:

1. $\mathbf{F}(\sigma z) = A(z)\mathbf{F}(z)$ for $A(z) \in \mathrm{GL}_n(\overline{\mathbb{Q}}(V))$ after adjoining finitely many inhomogeneous coordinates;
2. the four ordinary logarithms occur as $\overline{\mathbb{Q}}$-linear combinations of $\mathbf{F}(\alpha)$;
3. every degree-two functional relation among the corresponding four components is generated by functional linear relations;
4. specialization at $\alpha$ preserves that degree-two relation statement.

Only the determinant relation (43) needs to be lifted.

A useful relaxation is to allow an infinite ambient family but prove that the determinant lies in a finite invariant quotient. The root-alphabet identity then becomes input rather than an immediate obstruction: one would seek a finite quotient of the rooted parameter tree that retains the four special values and their quadratic tensor.

---

**Status:** The toric semi-invariant theorem, singular-subspace classification, quadratic descent proposition, Carlitz relation-ideal theorem, and Carlitz multiplier-rigidity theorem are **[proved here]**. Papanikolas's algebraic-independence theorem and Estienne's Mahler construction are **[imported theorem]**. The ordinary quadratic relation and finite operator module are **[open target]**. The root-alphabet calculation is an exact **[negative audit]**audit of one naive module, not a general impossibility theorem.

---

# 8 A differential logarithm lift, and why it specializes incorrectly

The root-alphabet obstruction in Section 7 concerns a difference operator. Ordinary logarithms also admit an elementary differential lift. At first sight it looks even better: the functions satisfy a finite rational differential system, and their functional algebraic relations can be determined completely. The difficulty is that this lift has *too few* functional relations. Multiplicative identities among the special values are created only at the specialization point.

## 8.1 Fixed-endpoint logarithm germs

For a rational number $q > 1$, put

$$c_q := 1 - \frac{1}{q}, \qquad \mathcal{L}_q(z) := -\log(1 - c_q z), \tag{47}$$

where the branch is the germ at $z = 0$. Then

$$\mathcal{L}_q(1) = \log q, \qquad \mathcal{L}_q'(z) = \frac{c_q}{1 - c_q z}. \tag{48}$$

Thus $(1, \mathcal{L}_{q_1}, \ldots, \mathcal{L}_{q_n})^t$ satisfies an inhomogeneous unipotent differential system with coefficients in $\mathbb{Q}(z)$, made homogeneous by adjoining the constant component. The point $z = 1$ is ordinary because $1/c_q = q/(q-1) > 1$.

The complete functional relation theorem is especially simple.

**Theorem 8.1** (Monodromy independence of the fixed-endpoint logarithms)**.** *Let $q_1, \dots, q_n > 1$ be distinct rational numbers. Then the germs*

$$\mathcal{L}_{q_1}(z), \dots, \mathcal{L}_{q_n}(z)$$

*are algebraically independent over $\mathbb{C}(z)$.*

*Proof.* The singular points $a_i = 1/c_{q_i}$ are distinct. Suppose that there is a nonzero polynomial relation

$$P(z, \mathcal{L}_{q_1}(z), \dots, \mathcal{L}_{q_n}(z)) = 0, \qquad P \in \mathbb{C}(z)[Y_1, \dots, Y_n],$$

and choose one of minimal total degree. Analytic continuation around a small loop enclosing $a_i$ and no other singularity replaces $\mathcal{L}_{q_i}$ by $\mathcal{L}_{q_i} + \omega$, where $\omega = \pm 2\pi i$, and leaves all other germs unchanged. Hence

$$\Delta_i P := P(z, Y_1, \dots, Y_i + \omega, \dots, Y_n) - P(z, Y_1, \dots, Y_n)$$

vanishes after substitution of the germs. If $P$ depends on $Y_i$, then $\Delta_i P$ is a nonzero polynomial relation of smaller total degree, a contradiction. Thus $P$ is independent of $Y_i$. Repeating this for every $i$ makes $P$ an element of $\mathbb{C}(z)$, which cannot vanish identically after substitution unless $P = 0$. ☐

If parameters are repeated, the full relation ideal is obtained by adjoining the obvious linear relations $Y_i - Y_j$ for equal $q_i = q_j$ and then applying the theorem to one representative of each distinct parameter.

## 8.2 The specialization defect already appears at integral controls

Theorem 8.1 does not transfer to $z = 1$. For example,

$$\mathcal{L}_4(1) - 2\mathcal{L}_2(1) = 0, \qquad \mathcal{L}_9(1) - 2\mathcal{L}_3(1) = 0, \tag{49}$$

while the three distinct germs $\mathcal{L}_2, \mathcal{L}_4, \mathcal{L}_9$ are algebraically independent over $\mathbb{C}(z)$. More generally, every multiplicative relation $\prod_i q_i^{m_i} = 1$ gives the value relation

$$\sum_i m_i \mathcal{L}_{q_i}(1) = 0, \tag{50}$$

although no corresponding functional relation exists when the $q_i$ are distinct.

For a candidate pair $(M, A)$ define

$$D_{M,A}(z) := \mathcal{L}_M(z)\mathcal{L}_3(z) - \mathcal{L}_A(z)\mathcal{L}_2(z). \tag{51}$$

If $2^x = M$ and $3^x = A$, then $D_{M,A}(1) = 0$. On the other hand, $D_{M,A}(z)$ is not the zero germ for any nonintegral candidate. Indeed, if $M \notin \{2, 3, A\}$ this follows from algebraic independence of the four germs. If $M = 3$, then $D_{M,A} = \mathcal{L}_3^2 - \mathcal{L}_A\mathcal{L}_2$ is nonzero by algebraic independence of the three distinct germs. The output $A$ is larger than both 3 and $M$ for a nonintegral candidate, so there are no other coincidences.

Even the valid control $x = 2$, $(M, A) = (4, 9)$, shows the correct mechanism: $D_{4,9}(1) = 0$ because the two linear relations in (49) generate the determinant relation at the value level, but neither linear relation lifts to the function level. Therefore no specialization theorem that simply preserves the functional relation ideal of the family (47) can prove the desired quadratic descent: it would reject the integral controls as well.

**Proposition 8.2** (Exact failure mode of the naive differential lift)**.** *The family $\{\mathcal{L}_q\}$ has a finite rational differential system and a completely rigid functional relation ideal, but evaluation at $z = 1$ is not relation-faithful even in degree one. Consequently it cannot be the function family in Target 7.10 without adding structure that lifts multiplication of the parameters before specialization.*

*Proof.* The differential system is (48); functional rigidity is Theorem 8.1; failure in degree one is (49). □

The missing structure is now clearer. A useful lift must remember that $\log(qr) = \log q + \log r$ at the *functional or motivic* level, not merely after evaluation. Kummer 1-motives do encode this functoriality, but their unconditional period theory controls linear relations, whereas the determinant lies in a tensor square. The Carlitz setting succeeds because the Frobenius difference module controls both the linear relation space and the full polynomial relation ideal.

**Research target 8.3** (Relation-faithful logarithm lift)**.** For the finitely generated subgroup of $\mathbb{Q}^{\times}_{>0}$ spanned by the prime divisors of $6MA$, construct analytic objects $\mathscr{K}_q$ with algebraic operator data and an algebraic specialization $\alpha$ such that:

1. $\mathscr{K}_q(\alpha) = \log q$;
2. every multiplicative relation among the $q$ lifts to the corresponding functional linear relation among the $\mathscr{K}_q$;
3. degree-two functional relations are generated by those lifted linear relations; and
4. specialization at $\alpha$ introduces no new degree-two relation.

For the conjecture only the determinant tensor is needed, not full algebraic independence in all degrees.

**Status:** The monodromy independence theorem and the specialization-defect proposition are **[proved here]**. They give a second exact no-go, independent of the Mahler root-alphabet obstruction. Target 8.3 is **[open target]**.

# 9 Rank-two logarithmic grids and the real additive threshold

Report III already established the carry-language lower bounds. The purpose of this section is not to repeat their construction. It is to place them against the precise July 2026 theorem of Harrison [7] and determine which kind of strengthening could actually contradict a counterexample.

## 9.1 The triangular smooth grid

For $R \geq 0$ define

$$\mathcal{G}_R := \{2^a 3^b : a, b \in \mathbb{N}_0,\ a + b \leq R\}. \tag{52}$$

Unique factorization gives

$$|\mathcal{G}_R| = \frac{(R+1)(R+2)}{2}, \qquad \mathcal{G}_R \subseteq \{1, 2, \dots, 3^R\}. \tag{53}$$

Its logarithm is the triangular rank-two generalized arithmetic progression

$$\log \mathcal{G}_R = \{a \log 2 + b \log 3 : a, b \geq 0,\ a + b \leq R\}.$$

Thus $\mathcal{G}_R$ is polynomially large in $R$ but only of size $\asymp (\log N)^2$ inside the ambient $N$-term progression with $N = 3^R$. It lies exactly at the intersection of two frontiers identified in Harrison's paper: sparse subsets and explicit rank or dimension dependence.

If $x$ were a counterexample and $M = 2^x$, $A = 3^x$, then

$$\mathcal{G}_R^{[x]} = \{M^a A^b : a, b \geq 0,\ a + b \leq R\} \subset \mathbb{Z}. \tag{54}$$

The same triangular exponent set survives, but the two geometric ratios become integers.

## 9.2 What Harrison's theorem gives on this grid

For fixed integers $1 \leq r \leq s$, put

$$\eta(r, s) := \max\{1, \min(r, s - 1)\}.$$

Harrison's Theorem 1.3 states that, for irrational $c$, the number of ordered non-permutation solutions of

$$\sum_{i=1}^{s} u_i^c = \sum_{j=1}^{r} v_j^c$$

with all variables in a subset of an $N$-term arithmetic progression is $O_s(|B|^{\eta(r,s)}(\log N)^{C_2(s)})$, where $C_2(s)$ is effective. The immediate specialization is the following.

**Corollary 9.1** (Fixed-arity Harrison bound on the smooth grid)**.** *Let $c$ be irrational and fix $1 \leq r \leq s$. The number of ordered non-permutation solutions with all variables in $\mathcal{G}_R$ is*

$$O_s\Big(R^{2\eta(r,s)+C_2(s)}\Big) \tag{55}$$

*as $R \to \infty$.*

*Proof.* Apply Harrison's theorem with $B = \mathcal{G}_R$ and the ambient progression $\{1, \ldots, 3^R\}$. Then $|B| \asymp R^2$ and $\log N = R \log 3$. □

This is a genuine restriction for every fixed arity. It is nevertheless compatible with all counterexample carries. A particularly transparent calibration comes from the two primitive moves. The two orders of expanding $(6t)^x$ give, for every $t \in \mathcal{G}_{R-1}$,

$$(M - 1)(3t)^x + A\, t^x = (A - 1)(2t)^x + M\, t^x. \tag{56}$$

Both sides have $M + A - 1$ terms, and the relation is non-permutational. Hence there are $\asymp R^2$ scaled instances at this fixed arity. Corollary 9.1 allows a much larger polynomial, so there is no tension.

### 9.3 From total energy to a fixed-target fiber

For a real exponent $c$, an integer $n \geq 1$, and a real target $T$, let

$$\mathcal{R}^{(c)}_{n,R}(T) := \#\left\{(m_g)_{g\in\mathcal{G}_R} \in \mathbb{N}_0^{\mathcal{G}_R} : \sum_g m_g = n, \quad \sum_g m_g g^c = T\right\}. \tag{57}$$

This counts unordered multiset representations. Harrison's ordered energy bound yields a fixed-target consequence, but its shape is too large in the arity needed here.

**Proposition 9.2** (Square-root fiber consequence)**.** *Fix an integer $n \geq 2$ and an irrational $c$. Then*

$$\max_T \mathcal{R}^{(c)}_{n,R}(T) \ll_n R^{n-1+C_2(n)/2} \tag{58}$$

*for $R \geq 2$.*

*Proof.* Choose one canonical ordering of each multiset in a target fiber. If the fiber contains $H$ distinct multisets, then ordered pairs of distinct canonical representatives give $H(H-1)$ non-permutation solutions of the equal-arity equation. Harrison's theorem with $r = s = n$ bounds their number by

$$O_n\big(|\mathcal{G}_R|^{n-1}(\log 3^R)^{C_2(n)}\big).$$

Use $|\mathcal{G}_R| \asymp R^2$ and take square roots. □

The dependence on fixed $n$ is essential. More importantly, even a hypothetical uniform version of the same *shape* would not be enough. If $n \asymp R$, the factor $R^{n-1}$ in (58) is

$$\exp\big((1+o(1))n\log R\big), \tag{59}$$

whereas the inherited carry fibers grow only exponentially in $n$. Making $C_2(n)$ and the implicit constant explicit cannot remove the $n\log R$ term. The missing estimate must exploit the fixed target and the boundary geometry, not merely count all equal sums.

### 9.4 The inherited exponential boundary fiber

For a hypothetical counterexample, Report III defines $\tau_x > 0$ by

$$\sum_{(a,b)\in\mathbb{N}_0^2\setminus\{(0,0)\}} \exp\big(-\tau_x(M^aA^b - 1)\big) = 1 \tag{60}$$

and proves that, along the appropriate residue class of arities, there are targets $T_n = M^{r_n}A^{s_n}$ with $r_n + s_n \leq n$ such that

$$\mathcal{R}^{(x)}_{n+1,n}(T_n) \gg_x \frac{\exp(\tau_x n)}{n^2}. \tag{61}$$

This statement is imported rather than reproved.

There is also a simpler mixed-direction calibration. Take $r = s = m$ primitive unit moves. The $\binom{2m}{m}$ possible orders give distinct multiset representations of the same target $M^mA^m$, all using

$$n_m = 1 + m(M + A - 2) \tag{62}$$

terms from $\mathcal{G}_{2m}^{[x]}$. Therefore

$$\liminf_{m\to\infty} \frac{1}{n_m} \log \mathcal{R}^{(x)}_{n_m,2m}(M^m A^m) \geq \frac{2\log 2}{M+A-2}. \tag{63}$$

This lower rate isolates the genuinely two-directional source of entropy: one primitive move in each direction has only one order, while interleaving the two creates the central binomial coefficient.

**Research target 9.3** (Boundary-conditioned mixed-fiber theorem)**.** Develop a rank-sensitive theorem for the triangular log-grid that bounds fixed-target fibers of arity $n \asymp R$ after isolating their character-balanced algebraic parts. It must do at least one of the following:

1. give, for the smooth boundary targets $(2^r 3^s)^c$, an upper exponential rate strictly below the mixed rate forced by (63); or
2. classify all positive-dimensional component signatures and prove that two independent primitive carry directions cannot coexist for irrational $c$ with integer coefficients.

A uniform extension of (55) with the same power of $|\mathcal{G}_R|$ is not sufficient.

Harrison's proof explains where such a theorem must differ. Transcendental rational points are controlled by o-minimal counting, while semi-algebraic curves are treated through functional transcendence and induction. The counterexample carries are character-balanced and therefore live in the algebraic part after a common dilation parameter is introduced. Their entropy can be hidden in the number of algebraic component signatures as arity grows. The required advance is thus a uniform classification or compression of those signatures, not a stronger estimate for the transcendental part alone.

---

**Status:** Corollary 9.1 and Proposition 9.2 are **[proved here]**from Harrison's imported theorem. The pressure lower bound is imported from Report III, and (63) is its elementary two-direction specialization. Target 9.3 is **[open target]**.

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## 10 Route triage and integral-control failure tests

A proposed obstruction is useful only if it survives every integral exponent. This section collects the exact failure tests produced by the new work. They are intentionally short: the point is to prevent future effort from rebuilding routes already falsified by a one-line control.

### 10.1 Same-prime congruence transfer fails at $x = 2$

Given an integer pair $(M, A)$ and a prime $p \nmid 6MA$, consider

$$\phi_p(r,s) = 2^r M^s \pmod p, \qquad \psi_p(r,s) = 3^r A^s \pmod p.$$

Over the reals, a common exponent identifies both expressions with powers having exponent $r + sx$. It is tempting to ask for the finite relation transfer

$$\ker \phi_p \subseteq \ker \psi_p. \tag{64}$$

This is not a necessary condition for the conjecture's valid solutions.

**Proposition 10.1** (Integral-control counterexample to same-prime transfer). *For $x = 2$, $(M, A) = (4, 9)$, and $p = 7$, the vector $(3, 0)$ belongs to $\ker \phi_7$ but not to $\ker \psi_7$.*

*Proof.* One has $2^3 \equiv 1 \pmod 7$, whereas $3^3 \equiv 27 \equiv 6 \pmod 7$. □

Thus no argument may infer (64) directly from the real common exponent, whether at one prime or uniformly at every prime. A viable local-to-global theorem must use a different-prime modulus chosen to dominate the target order, an almost-everywhere support hypothesis proved by independent arithmetic input, or a richer joint Kummer condition. The real identity alone supplies none of these.

### 10.2 A compact failure matrix

| Route | Exact obstruction | What remains viable |
|---|---|---|
| Fixed finite monomial library | Theorem 5.1: every fixed support becomes extinct beyond an effective block. | Supports whose size, frequency diameter, or near-resonance quality grows with $L$. |
| Pure toric scaling | Theorem 7.1: polynomial semi-invariants of $(X, Y) \mapsto (2X, 3Y)$ are monomials. | A nondiagonal operator, a quotient of an infinite module, or a tensor construction. |
| Fixed-endpoint differential logs | Theorem 8.1 and (49): specialization creates even the control relations. | A relation-faithful Kummer or motivic lift in which multiplication is functional before evaluation. |
| Naive Mahler pullback | Equation (46): repeated pullback creates an infinite root alphabet. | A finite invariant quotient of the root tree or a different operator family. |
| Dense near-curve support | Equation (38): the predicted total support capacity is never $o(L)$. | Sparse support-restricted matrices and certificate covers. |
| Fixed-arity additive energy | Equation (59): its natural growing-arity continuation costs $\exp(\Theta(n \log R))$. | Boundary-conditioned fixed-target classification of algebraic component signatures. |
| Same-prime congruence transfer | Proposition 10.1: it fails for $x = 2$ modulo 7. | Output-specific Kummer information or a modulus-changing continuity theorem. |

Three common mistakes are thereby separated.

1. **A search algorithm is not a proof criterion.** Faster enumeration of a block is valuable, but the proof objective is the zero-capacity inequality of Theorem 4.7.
2. **A functional relation theorem is not a value theorem.** One must audit which relations are created at the specialization point; integral controls are the fastest test.
3. **A local shadow is not an archimedean transfer.** The equality of real logarithmic ratios does not by itself impose compatible finite kernels.

**Status:** Proposition 10.1 is **[proved here]**. Every row of the failure matrix points to a theorem or exact calculation in this report. The surviving alternatives are research directions, not claims of existence.

## 11 Formalization architecture and certificate checking

The new arguments divide naturally into a small unconditional core, external theorem interfaces, and computational certificates. This is preferable to formalizing one monolithic proof attempt: failed searches remain untrusted, while every accepted row has a short kernel-checkable meaning.

### 11.1 Dependency map

| Component | Formal content | Dependencies |
|---|---|---|
| Rational exponent lemma | If $2^{a/b} \in \mathbb{Z}$ in lowest terms, then $b = 1$. | Integer valuations only. |
| Block pressure | Construction $n_k = L - \lfloor kx \rfloor$ and distinctness of $\{kx\}$. | Ordered fields, floor arithmetic, irrationality. |
| Dyadic normalization | $(2^L z)^\theta = 3^L z^\theta$. | Real logarithm identities. |
| Integer locking | An integer of absolute value $< 1$ is zero. | Discrete ordered ring. |
| Exponential zero budget | A $T$-term real exponential sum has at most $T - 1$ zeros. | Rolle's theorem and induction. |
| Curve zero budget | Distinctness of $i + j\theta$ and substitution $z = e^t$. | Irrationality of $\theta$ plus the preceding item. |
| Cover capacity | Assign each exact point to a covering certificate and sum zero budgets. | Finite sets and cardinal inequalities. |
| Support conditioning | Explicit inverse-Vandermonde row bound. | Polynomial coefficient bounds, mean-value theorem, finite matrices. |
| Toric semi-invariants | Compare monomial coefficients under $P(2X, 3Y) = cP(X, Y)$. | Unique factorization of monomials. |
| Singular matrix plane | A two-dimensional all-singular subspace with surjective first row has a fixed row multiplier. | $2 \times 2$ determinants and linear algebra. |
| Monodromy independence | Minimal polynomial relation plus translation monodromy. | Analytic continuation around distinct logarithmic branch points. |
| Same-prime control failure | $2^3 \equiv 1$, $3^3 \equiv 6 \pmod 7$. | Decidable modular arithmetic. |

The following results should enter as named interfaces rather than be reproved in the first formalization pass:

**Baker interface.** A polynomial lower bound for nonzero $a + b\theta$.

**Near-curve interface.** The Brisebarre–Hanrot rounded Chebyshev norm theorem.

**Carlitz interface.** Papanikolas's relation-ideal theorem for algebraic Carlitz logarithms.

**Additive interface.** Harrison's fixed-arity non-permutation solution bound.

Each interface is used in one sharply delimited theorem. Removing it leaves the rest of the file executable and exposes the exact unformalized assumption.

### 11.2 Suggested Lean theorem signatures

The names below are schematic but sufficiently precise to guide implementation.

```
theorem rational_exponent_integral
    {a b M : Int} (hb : 0 < b) (hcop : IsCoprime a b)
    (hpow : (2 : Real) ^ ((a : Real) / b) = M) : b = 1

theorem block_pressure
    (hx : Irrational x) (hsol2 : (2 : Real)^x = M)
    (hsol3 : (3 : Real)^x = A) :
    card (blockPoints x L) >= ceil ((L + 1) / x)

theorem exp_sum_zero_count
    (hfreq : StrictMono lambda) (hcoeff : forall i, coeff i != 0) :
    card (zerosOn (fun t => sum i, coeff i * exp (lambda i * t)) I)
      <= T - 1

theorem curve_polynomial_zero_count
    (htheta : Irrational theta) (hP : P != 0) :
    card {z in I | P(z, z^theta) = 0} <= supportCard P - 1

theorem certified_cover_capacity
    (hcover : Covers unitInterval certs)
    (hsubunit : forall C in certs, IsSubunitCertificate L C) :
    card (integerCurvePoints L) <= sum C in certs, supportCard C.poly - 1

theorem support_conditioning_barrier
    (hsubunit : supNormOnBlock theta L P < 1) :
    L < conditioningBudget theta P.support

theorem toric_semiInvariant_is_monomial
    (h : renameScale 2 3 P = c * P) : IsMonomial P

theorem singular_plane_fixed_multiplier
    (hsing : forall X in W, det X = 0)
    (hsurj : Surjective firstRowOnW) :
    exists q, forall X in W, secondRow X = q * firstRow X
```

The block-pressure theorem should represent points by their real exponent $s_k$ first and only then map to integer pairs. This avoids proving accidental injectivity through transcendental coordinates. The curve-zero theorem should normalize the support into an ordered list of distinct frequencies and reuse one generic exponential-sum theorem.

### 11.3 A minimal certificate object

An untrusted search program should emit records of the following logical form.

```
structure NearCurveCertificate where
  block       : Nat
  left right  : Rat
  hInterval   : left < right
  support     : Array (Nat x Nat)
  coeff       : Array Int
  N1 N2       : Nat
  rho1 rho2   : Rat
  thetaBox    : Rat x Rat
  chebBox     : Array (Rat x Rat)
  tailBox     : Array (Rat x Rat)
  roundedA    : Matrix Int
  scaleExp    : Nat
  normSqUpper : Rat
  hNorm       : normSqUpper < 1 / (support.size + N1*N2)^2
  hAnalytic   : CertifiedAnalyticEnvelope
```

The checker performs no lattice reduction. It verifies array dimensions, support uniqueness, rational interval containment, directed-rounding enclosures, the exact squared-norm inequality, and the analytic tail theorem. The resulting proposition is

$$\forall z \in [a, b], \quad |P(2^L z, 3^L z^\theta)| < 1.$$

A cover certificate is a finite array of these records plus an exact rational proof that the intervals cover $[1, 2]$. Its final output is the integer

$$\mathrm{Cap} = \sum_C (|\operatorname{supp} P_C| - 1),$$

which is compared directly with the orbit-pressure lower bound.

### 11.4 Trusted and untrusted layers

| Layer | Responsibility |
|---|---|
| Untrusted search | Support enumeration, Chebyshev sampling, precision choice, LLL or BKZ, interval subdivision, and candidate ranking. |
| Exact checker | Integer matrix arithmetic, rational enclosures, norm inequalities, support capacity, and interval cover. |
| Analysis library | Bernstein ellipse bounds, Chebyshev tails, exponential zero theorem, and continuity facts. |
| External theorem interface | Baker separation and, in later branches, Papanikolas or Harrison. |
| Final kernel theorem | If a certified cover has capacity smaller than the forced counterexample point count, then no noninteger solution exists. |

This architecture permits formal progress before the analytic search succeeds. In particular, the finite checker, block-pressure theorem, zero-capacity theorem, and the degree-three pilot certificate format can be completed independently.

**Status:** The dependency architecture is a formalization plan. The theorem statements mirror results already proved in this report, but no Lean source is claimed to have been compiled. External theorems are explicitly isolated.

# 12 Prioritized research program

The report has reduced several broad ideas to executable tests. The priorities below are ordered by the ratio of mathematical payoff to missing infrastructure, not by conceptual prestige.

## 12.1 Priority A: certified sparse near-curve search

This is the most direct route because every successful record has unconditional proof content. The first implementation should follow the loop below.

1. **Enumerate support families.** Use total degree $d \asymp \sqrt{L/\log L}$ and support size at most $CL/\log L$. Include anisotropic lower sets, near-resonant shells around convergents to $\theta$, and a small low-frequency core.
2. **Apply the conditioning prefilter.** Compute $\mathfrak{B}(S)$ exactly enough to reject every support with $\mathfrak{B}(S) \le L$. This costs essentially nothing compared with lattice reduction and prevents searches forbidden by Theorem 5.1.
3. **Build the support-restricted Chebyshev matrix.** Use the exact dyadic row scales $2^{Li}3^{Lj}$, cached analytic envelopes for the interval shape, and directed rounding.
4. **Reduce and certify.** LLL or BKZ proposes rows. The independent checker verifies (30). A row that is small only on sample nodes is discarded.
5. **Subdivide only on failure.** If no row certifies an interval, bisect it. Charge the actual support of every accepted row and monitor

$$\Gamma_L := \frac{1}{L}\sum_C (|\operatorname{supp} P_C| - 1).$$

   The asymptotic goal is $\Gamma_L \to 0$, not merely a fast running time.

The first milestones are deliberately finite:

**A0.** Convert the degree-three rows for $L \le 5$ into rigorous interval certificates or prove that those particular rows fail between grid points.

**A1.** Reproduce the dense Brisebarre–Hanrot certificate on a small block with an independent checker, then delete rows to verify Proposition 6.2 computationally.

**A2.** For $10 \le L \le 100$, record the best certified values of support count, degree, conditioning budget, interval count, and $\Gamma_L$.

**A3.** Search for a family whose empirical capacity grows more slowly than $L$ and then replace the observed pattern by a uniform analytic construction.

A negative outcome is still informative. If the minimal certified capacity stabilizes above a positive multiple of $L$ while support budgets approach the conditioning floor, the one-polynomial cover strategy is likely at its natural limit and the search should move to operator or additive structure rather than larger blind lattices.

### 12.2 Priority B: analytic design of near-resonant supports

The conditioning budget shows why convergents to $\theta$ are relevant. If $p/q$ approximates $\theta$, frequency pairs separated by $(p,-q)$ have gap $|p-q\theta|$, increasing $\mathfrak{B}(S)$. A pure chain in this direction is not sufficient: it reduces to a gap-power construction already ruled out in Report III. The viable design is a *two-dimensional* support with a small core and several translated near-resonant pairs.

A concrete family to analyze is

$$S(d,p,q,h) := \{(i,j) : 0 \le i+j \le d_0\} \cup \bigcup_{r=1}^{h} \{(i_r,j_r),(i_r+p,j_r-q)\}, \tag{65}$$

with all coordinates nonnegative and total degree at most $d$. The core controls low-order Chebyshev behavior; the shell supplies small frequency gaps without collapsing the whole polynomial to one variable. The research tasks are:

1. derive a determinant or singular-value bound sharper than the worst-case Theorem 5.1 for (65);
2. estimate the Brisebarre–Hanrot tail weights monomial by monomial;
3. determine whether $h \asymp L/\log L$ can be supported at $d \asymp \sqrt{L/\log L}$; and
4. prove that any successful short vector is not forced into a line-supported factor.

The success criterion is an explicit support family and coefficient-selection theorem, not a list of isolated LLL rows.

### 12.3 Priority C: relation-faithful operator or motivic lift

The Carlitz theorem proves that the desired multiplier rigidity is natural in a category with a strong difference operator. The two ordinary-log audits identify the specifications:

1. multiplication of algebraic parameters must produce functional linear relations;
2. the degree-two determinant must lie in a finite tensor or quotient module;
3. the operator must not diagonalize monomials as the toric scaling does;
4. specialization must preserve the degree-two relation ideal while allowing the known linear relations of integral controls.

The next precise calculation is to express

$$\log M \otimes \log 3 - \log A \otimes \log 2$$

as a period tensor of the direct sum of the Kummer 1-motives attached to $2, 3, M, A$, and compute the smallest Tannakian subquotient containing it. One should then ask for a *single-tensor* period statement on that subquotient, rather than invoke the full Grothendieck–Andre or Schanuel conjecture. The characteristic-$p$ proof in Theorem 7.5 supplies the model answer that the relation ideal should have.

A proposed special-function family must be tested immediately on $(M, A) = (4, 9)$ and $(8, 27)$. If the value-linear relations do not lift functionally, the family is rejected before any algebraic-independence argument is attempted.

### 12.4 Priority D: growing-arity algebraic-part classification

For Harrison's route, improving the exponent of the transcendental point count is not the first task. The carry points already lie in algebraic parts. The useful program is:

1. parameterize character-balanced semi-algebraic components for equal-sum equations on the triangular log-grid;
2. encode a component by a finite partition and a lattice of multiplicative characters;
3. bound the number of component signatures meeting a fixed boundary target when arity is $O(R)$; and
4. isolate the signature operation corresponding to interchanging the two primitive carry directions.

A theorem with total count $|\mathcal{G}_R|^{O(n)}$ should be regarded as a failure, because (59) remains. The target is exponential in $n$ with a controlled base, or a structural statement that forbids two independent integer carry directions for irrational $c$.

### 12.5 Priority E: local arithmetic only with an explicit transfer lemma

The existing Kummer densities and support theorems are powerful once a local hypothesis is available. Proposition 10.1 shows that the real common exponent does not provide the simplest one. New local work should therefore begin by stating an archimedean-to-local transfer lemma in full, including its quantifiers over primes and depth. Unless that lemma survives all integral controls and yields more than a same-prime kernel inclusion, no further Chebotarev computation is warranted.

### 12.6 Decision table

| Track | Continue when | Stop or redesign when |
|---|---|---|
| Sparse certificates | Certified $\Gamma_L$ decreases and surviving supports have $\mathfrak{B}(S) > L$ by a growing margin. | Only sample-small rows appear, or certified capacity remains $\gg L$ near the conditioning floor. |
| Near-resonant design | A two-dimensional core-shell support improves certified singular values without line factorization. | Every good vector reduces to a one-variable gap power. |

| Track | Continue when | Stop or redesign when |
|---|---|---|
| Operator lift | Integral-control linear relations lift before specialization and the determinant lies in a finite tensor quotient. | The family is diagonal, has an uncontrolled infinite alphabet, or creates control relations only at the value. |
| Additive classification | The number of boundary component signatures has exponential rate independent of $\log R$. | The estimate retains a factor $R^{cn}$ or counts only the transcendental part. |
| Local arithmetic | A proved transfer supplies compatible Kummer data for a positive-density or growing-depth prime family. | The hypothesis is simply inferred from equality of real logarithmic ratios. |

## 13 Conclusion

No proof of the Alaoglu–Erdős conjecture has been obtained. The new progress is nevertheless concrete and substantially narrows three routes.

On the auxiliary-polynomial side, the block normalization is exact and scale-free: $f(z) = z^{\theta}$, $u = 2^L$, $v = 3^L$. A counterexample would place linearly many exact points in every block, while a $T$-term polynomial can vanish at at most $T - 1$ positive curve points. This yields the first proof criterion of the report:

$$\text{a certified cover of total support } o(L) \text{ proves the conjecture.}$$

The unconditional inverse-Vandermonde argument shows that a global certificate must have degree at least $\gg \sqrt{L/\log L}$; the counterexample pressure requires $\gg \sqrt{L}$. The remaining construction window is logarithmic. Brisebarre–Hanrot's 2026 algorithm can be restricted rigorously to arbitrary sparse supports, turning this criterion into an executable search and formal-certificate program.

On the transcendence side, Papanikolas's theorem gives an exact positive-characteristic model: two independent algebraic Carlitz logarithms cannot be multiplied by a common scalar and remain algebraic Carlitz logarithms unless the scalar is rational over the function field. Linear algebra shows that the ordinary conjecture would follow from the much narrower statement that one quadratic determinant relation among four logarithms is generated by their rational linear relations. Pure toric scaling cannot provide the operator. The naive Mahler family has an infinite root alphabet, while the naive differential family has the opposite defect: its germs are algebraically independent, but specialization creates even $\log 4 = 2 \log 2$. A successful lift must therefore be relation-faithful before specialization.

On the additive side, Harrison's fixed-arity theorem gives polynomial bounds on the triangular smooth grid. Passing from total energy to a target fiber and then allowing arity proportional to $R$ introduces an unavoidable $\exp(\Theta(n \log R))$ wall, far above the inherited carry entropy. The required theorem must condition on the smooth boundary target and classify the character-balanced algebraic components; improving only the transcendental point count cannot succeed.

The most immediate opportunity is the sparse near-curve program because its output is self-certifying and its failure modes are quantitative. The deepest conceptual opportunity is the single-tensor operator target suggested by the Carlitz analogue. The additive route becomes competitive only after its algebraic parts are counted with growing-arity, rank-sensitive constants. These three directions are independent enough that progress in any one would change the status of the conjecture, and precise enough that future work can be judged against explicit thresholds rather than qualitative plausibility.

**Sharp remaining bottlenecks.**

1. Construct a certified dyadic cover with total monomial capacity $o(L)$.
2. Prove degree-two relation descent for the determinant tensor of $\log 2, \log 3, \log M, \log A$.
3. Prove a boundary-conditioned growing-arity classification whose exponential rate beats the mixed carry rate.

Any one of these closes the conjecture; none is claimed here.

# A Logical dependency map and certificate formats

This appendix separates the report's unconditional mathematics from the three conditional closure routes. Its purpose is practical: a proof assistant, an independent reader, or a computational team should be able to replace one module without silently changing another.

## A.1 The dyadic route as a short implication chain

For a fixed real $x$ satisfying $M = 2^x \in \mathbb{Z}$ and $A = 3^x \in \mathbb{Z}$, the dyadic argument has the following dependency graph:

$$\begin{gathered} x \notin \mathbb{Z} \Longrightarrow x \notin \mathbb{Q} \Longrightarrow \#\mathcal{B}_L \geq \lceil (L+1)/x \rceil \\ \Downarrow \\ \text{subunit polynomial on a covering interval} \Longrightarrow \text{all exact integer points there are zeros} \\ \Downarrow \\ P(X, X^\theta) = \sum c_{ij} X^{i+j\theta} \Longrightarrow \#\{P = 0\} \leq T(P) - 1 \\ \Downarrow \\ \mathrm{Cap}_L \geq \lceil (L+1)/x \rceil . \end{gathered}$$

Every arrow in this diagram is proved in Sections 2–4. No transcendence theorem beyond the elementary irrationality of $\log_2 3$ is used. The sole missing input for a proof is the construction of covers whose capacity is $o(L)$.

A machine-checkable global certificate for block $L$ may therefore consist of records

$$\mathfrak{C}_{L,r} = (a_r, b_r, S_r, (c_{ij})_{(i,j)\in S_r}, \varepsilon_r), \tag{66}$$

where all interval endpoints and error bounds are rational, together with proofs of

$$[1,2] \subseteq \bigcup_r [a_r, b_r], \tag{67}$$

$$\sup_{z\in[a_r,b_r]} \left| \sum_{(i,j)\in S_r} c_{ij} 2^{Li} 3^{Lj} z^{i+j\theta} \right| \leq 1 - \varepsilon_r, \qquad \varepsilon_r > 0, \tag{68}$$

$$\sum_r (|S_r| - 1) < L/C \tag{69}$$

for a sequence of constants $C \to \infty$. The analytic inequality may be certified by rational Chebyshev enclosures, Bernstein ellipses, Taylor models, or any other validated method. The proof kernel need not trust the lattice reduction that proposed the coefficients.

### A.2 A support-conditioning prefilter

Before attempting interval certification, a support $S$ can be rejected whenever $L \geq \mathfrak{B}(S)$. A rational implementation should not evaluate $\theta$ as an unproved floating-point constant. Instead, it may carry directed rational intervals

$$\underline{\theta} < \frac{\log 3}{\log 2} < \overline{\theta}$$

obtained from certified logarithm bounds. From these one obtains:

1. an upper bound for every frequency $i + j\theta$ and hence for $\Lambda(S)$;
2. a lower bound for every nonzero difference $(i - i') + (j - j')\theta$ and hence for $\delta(S)$;
3. an outward-rounded upper bound for $\mathfrak{B}(S)$.

For large degree, Baker–Wüstholz or Matveev supplies a symbolic lower bound for the third item. For moderate supports, direct certified evaluation of the finitely many frequency differences is usually much sharper. Thus the theoretical degree barrier and the practical pruning test use the same theorem but should not use the same numerical constants.

### A.3 The operator route as relation-ideal descent

The function-field part has an equally short logical shape. For Carlitz logarithms, Papanikolas gives

$$k\text{-linear independence} \Longrightarrow \overline{k}\text{-algebraic independence.}$$

Choosing a basis converts this into the statement that every polynomial relation is generated by linear relations. Applying that statement to the determinant $X_1X_4 - X_2X_3$ and invoking Lemma 7.2 forces the common multiplier to lie in $k$.

For ordinary logarithms, no full algebraic-independence theorem is required. A certificate of the needed relation descent would be an identity

$$X_1X_4 - X_2X_3 = \sum_{r=1}^{m} L_r(X_1, X_2, X_3, X_4) Q_r(X_1, X_2, X_3, X_4), \tag{70}$$

where the $L_r$ are rational linear forms known to vanish at $(\log 2, \log 3, \log M, \log A)$ and the $Q_r$ are linear forms over $\overline{\mathbb{Q}}$. Since the left side has degree two, only the degree-one part of each $Q_r$ matters. Hence a successful operator theorem ultimately produces a finite matrix identity, even if its proof uses a much larger functional system.

The differential audit in Section 8 imposes a necessary test on every proposed lift: it must reproduce the valid integral-control relation

$$\log 4 - 2\log 2 = 0$$

at the functional level, or prove by a regular-specialization theorem that this relation descends from the chosen functional relation ideal. Functional algebraic independence by itself is not an advantage if specialization creates new relations.

### A.4 The additive route and the required theorem shape

Let $\mathcal{G}_R = \{2^a 3^b : a + b \le R\}$. A theorem useful for the conjecture must distinguish three quantifiers that are harmless at fixed arity but decisive when $n \asymp R$:

1. the exponent $x$ is a fixed irrational number satisfying the two-base integrality hypothesis;
2. the arity $n$ grows linearly with $R$;
3. the target is not arbitrary but belongs to the smooth boundary grid $\mathcal{G}^{[x]}_{O(R)}$ and is accompanied by character-balance data inherited from the two generators.

A prototype conclusion strong enough to interact with the carry lower bound would have the form

$$\max_{T \in \mathcal{G}^{[x]}_{cR}} \#\{(g_1, \dots, g_n) \in \mathcal{G}_R^n : g_1^x + \cdots + g_n^x = T\} \le \exp\big((\sigma + o(1))n\big), \tag{71}$$

with $\sigma$ strictly smaller than the independently certified carry entropy. Any estimate with an uncontrolled factor $R^{cn}$ has logarithm $cn \log R$ and cannot meet this threshold. Estimate (71) is not asserted; it records the exact asymptotic shape that a future higher-rank theorem must achieve.

### A.5 Closure conditions and independence of the routes

The three closure modules are logically independent:

| Route | Sufficient new input |
|---|---|
| Dyadic auxiliary polynomials | Certified covers with $\sum_r (\lvert S_{L,r}\rvert - 1) = o(L)$ along an unbounded sequence of block indices. |
| Operator/transcendence | Degree-two relation descent for the determinant tensor of the four ordinary logarithms, uniformly for every putative solution. |
| Additive classification | A boundary-conditioned growing-arity fiber upper bound whose exponential rate is strictly below the imported carry-fiber lower rate. |

A proof from the first route would not require the operator or additive targets. A proof from the second would not require any computational certificate. A proof from the third would use the carry construction from Report III but not the near-curve algorithm. This modularity is useful: a failed experiment in one track does not weaken the exact reductions in the others.

## B Reproducibility script

The following Python script reproduces the numerical constants in Tables 3, 4, and 5, including the explicit polynomial (27). It uses arbitrary-precision evaluation for reconnaissance and exact integer arithmetic for LLL. As emphasized in the main text, the sampled supremum is not an interval certificate.

```
"""Reproduce the numerical data in Continuation Report IV.

```

```
Requirements:
    Python 3.11 or later
    mpmath
    sympy

The LLL lattice is exact.  The final supremum is sampled on 2001 points
and is therefore diagnostic, not a proof of a uniform inequality.
"""

from __future__ import annotations

from dataclasses import dataclass
from typing import Iterable

import mpmath as mp
from sympy import Matrix

mp.mp.dps = 100
THETA = mp.log(3) / mp.log(2)


def chebyshev_table() -> None:
    """Print the rho=5 error and conservative block-scale table."""
    constant = mp.power(mp.mpf("2.8"), THETA)
    print("Chebyshev diagnostics (rho=5)")
    print("degree       E_n                    B")
    for degree in (8, 12, 20, 28, 36, 48):
        error = constant * mp.power(5, -degree)
        block_scale = mp.power(1 / (8 * error), 1 / THETA)
        print(
            f"{degree:>6d}  "
            f"{mp.nstr(error, 12):>20s}  "
            f"{mp.nstr(block_scale, 12):>20s}"
        )


def bh_exponent_table() -> None:
    """Print the dense-support exponents in the dyadic variables."""
    print("\nBrisebarre--Hanrot exponent translation")
    print("d       N_d            a_d          kappa_d")
    for degree in (4, 6, 8, 10, 12, 16, 20, 24, 32, 40, 64):
        dimension = (degree + 1) * (degree + 2) // 2
        a_d = mp.mpf(2 * degree) / (3 * dimension)
        kappa_d = (1 + THETA) * a_d
        print(
            f"{degree:>2d}  {dimension:>8d}  "
            f"{float(a_d):>13.6f}  {float(kappa_d):>13.6f}"
        )


@dataclass(frozen=True)
class PilotResult:
    block: int
    sampled_supremum: mp.mpf
    coefficients: tuple[int, ...]


def total_degree_monomials(degree: int) -> list[tuple[int, int]]:
    """Order monomials by Y-degree, then X-degree."""
    return [
```

```
        (i, j)
        for j in range(degree + 1)
        for i in range(degree - j + 1)
    ]


def evaluate_polynomial(
    coefficients: Iterable[int],
    monomials: list[tuple[int, int]],
    block: int,
    z: mp.mpf,
) -> mp.mpf:
    """Evaluate P(2^L z, 3^L z^theta) at high precision."""
    x_value = mp.power(2, block) * z
    y_value = mp.power(3, block) * mp.power(z, THETA)
    total = mp.mpf("0")
    for coefficient, (i, j) in zip(coefficients, monomials):
        total += (
            coefficient
            * mp.power(x_value, i)
            * mp.power(y_value, j)
        )
    return total


def lll_pilot(block: int, degree: int = 3) -> PilotResult:
    """Run the exact sample lattice used in the report."""
    monomials = total_degree_monomials(degree)
    dimension = len(monomials)
    coefficient_scale = 10**8
    evaluation_scale = 10**30

    nodes = [
        mp.mpf("1.5")
        + mp.mpf("0.5")
        * mp.cos((k + mp.mpf("0.5")) * mp.pi / dimension)
        for k in range(dimension)
    ]

    rows: list[list[int]] = []
    for row_index, (i, j) in enumerate(monomials):
        row = [0] * dimension
        row[row_index] = coefficient_scale
        for z in nodes:
            value = (
                mp.power(2, block * i)
                * mp.power(3, block * j)
                * mp.power(z, i + j * THETA)
            )
            row.append(int(mp.nint(evaluation_scale * value)))
        rows.append(row)

    reduced = Matrix(rows).lll()
    best_supremum: mp.mpf | None = None
    best_coefficients: tuple[int, ...] | None = None

    for row_index in range(reduced.rows):
        first_coordinates = [
            int(reduced[row_index, column])
            for column in range(dimension)
```

```
        ]
        if any(value % coefficient_scale for value in first_coordinates):
            raise ArithmeticError("LLL row left the coefficient sublattice")
        coefficients = tuple(
            value // coefficient_scale for value in first_coordinates
        )

        # Exclude the trivial constant polynomial, whose sampled norm is one.
        if all(value == 0 for value in coefficients[1:]):
            continue

        sampled_supremum = mp.mpf("0")
        for grid_index in range(2001):
            z = mp.mpf(1) + mp.mpf(grid_index) / 2000
            sampled_supremum = max(
                sampled_supremum,
                abs(
                    evaluate_polynomial(
                        coefficients,
                        monomials,
                        block,
                        z,
                    )
                ),
            )

        if (
            best_supremum is None
            or sampled_supremum < best_supremum
        ):
            best_supremum = sampled_supremum
            best_coefficients = coefficients

    if best_supremum is None or best_coefficients is None:
        raise RuntimeError("No nonconstant reduced row was found")

    return PilotResult(block, best_supremum, best_coefficients)


def format_polynomial(
    coefficients: tuple[int, ...],
    monomials: list[tuple[int, int]],
) -> str:
    """Return a compact machine-readable polynomial string."""
    terms = []
    for coefficient, (i, j) in zip(coefficients, monomials):
        if coefficient == 0:
            continue
        terms.append(f"({coefficient})*X^{i}*Y^{j}")
    return " + ".join(terms)


def pilot_table() -> None:
    print("\nDegree-three LLL pilot")
    monomials = total_degree_monomials(3)
    for block in range(1, 7):
        result = lll_pilot(block)
        print(
            f"L={block}: "
            f"sampled sup={mp.nstr(result.sampled_supremum, 15)}"
```

```
        )
        if block == 5:
            print("L=5 coefficients:", result.coefficients)
            print("L=5 polynomial:")
            print(format_polynomial(result.coefficients, monomials))


if __name__ == "__main__":
    print("theta =", mp.nstr(THETA, 50))
    chebyshev_table()
    bh_exponent_table()
    pilot_table()
```

Listing 1: Reproduction of the numerical tables and the degree-three LLL pilot.

On the versions used for this report, the six sampled suprema are

$$6.9932152582 \times 10^{-5}, \quad 7.9924631253 \times 10^{-4}, \quad 6.3015062341 \times 10^{-3},$$

$$5.8264974254 \times 10^{-2}, \quad 5.5874173100 \times 10^{-1}, \quad 4.0991943672.$$

Exact agreement of the coefficient tuple for $L = 5$ is the most useful implementation check, because it tests the monomial ordering, lattice orientation, and scaling simultaneously.

## C Literature and search audit

The literature search was intentionally narrow: only results that alter one of the report's three quantitative routes were imported. The following table records the role of each recent source and what is *not* inferred from it.

| Source | Imported content | Boundary of use |
|---|---|---|
| Brisebarre–Hanrot (June 2026) | A rigorously checked short-vector condition yielding an integer polynomial uniformly smaller than one near an analytic curve; asymptotic interval and degree parameters for the dense triangular support. | The report does not assume their heuristic coprimality/elimination step. The sparse-support extension used here follows by deleting rows from the same matrix and rechecking the norm implication. |
| Estienne (March 2026) | A Mahler-method proof of algebraic independence for linearly independent Carlitz logarithms, exhibiting a finite semilinear functional system and a specialization theorem. | No characteristic-zero analogue is claimed. The paper is used to identify the operator and specialization features that a new argument would need. |
| Harrison (December 2025; revised July 2026) | Fixed-arity bounds for additive relations among irrational powers of sets lying in generalized arithmetic progressions, including logarithmic losses depending on the ambient scale. | The constants are not uniform in arity, and the theorem does not classify the boundary-conditioned algebraic components required here. |

| Source | Imported content | Boundary of use |
|---|---|---|
| Papanikolas (2008) | Algebraic independence of $k$-linearly independent Carlitz logarithms of algebraic points. | The multiplier-rigidity theorem is a corollary proved in this report; it is a positive-characteristic benchmark, not a proof of the ordinary conjecture. |
| Baker–Wüstholz; Matveev | Effective lower bounds for nonzero linear forms in logarithms of fixed algebraic numbers. | Only the existence of an effective polynomial separation $\lvert a + b\theta\rvert \gg H^{-C}$ is used; no optimized numerical constant is asserted. |

The search also checked the competitive claims in the prompt. Those sources establish useful context, but none contains a disclosed approach to the Alaoglu–Erdős conjecture. In particular, no inference about the rival team's private Lean development enters any proof or target in this report.

## References


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